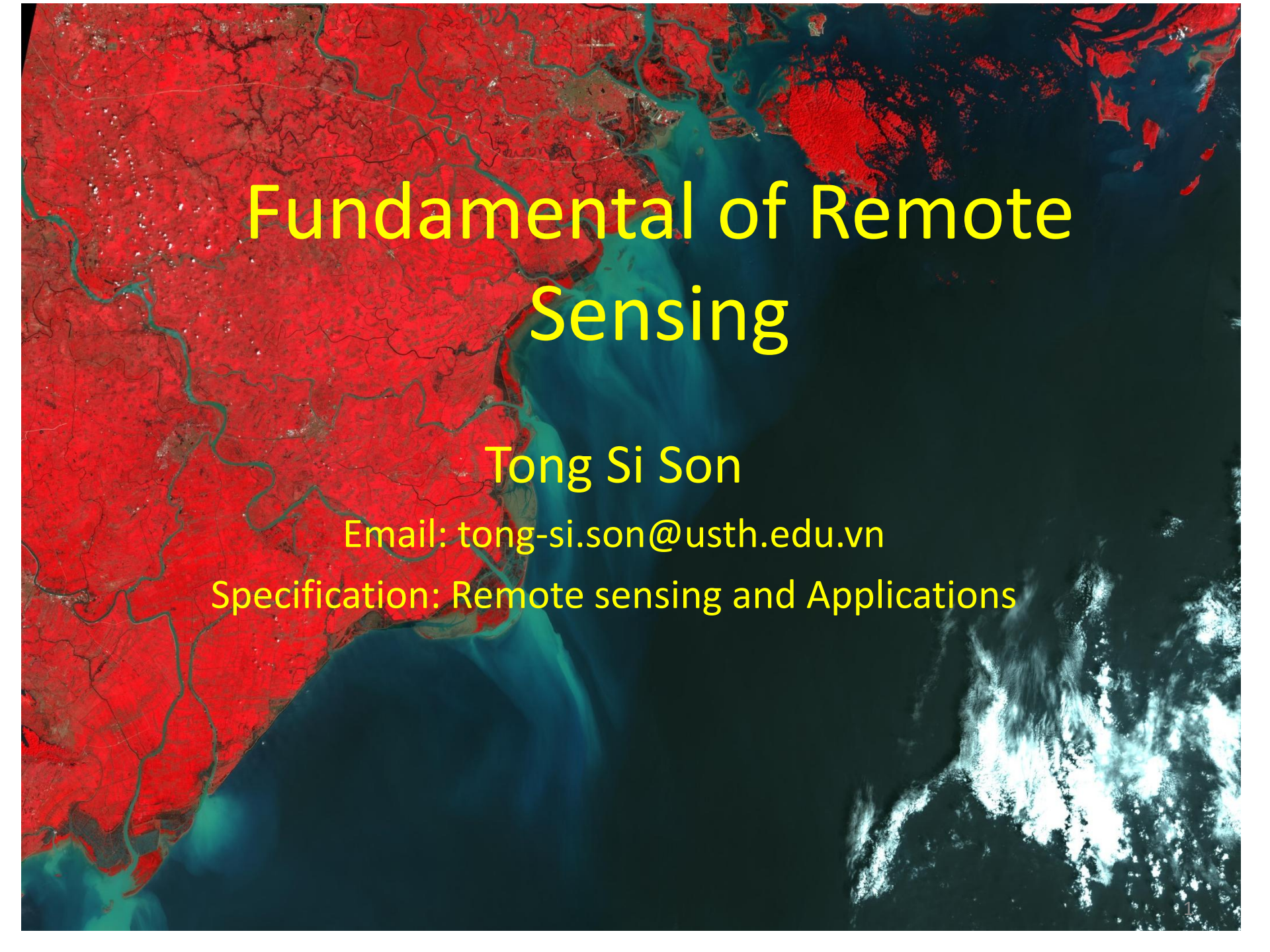


A satellite image of a coastal region. The land is colored in a vibrant red, showing intricate patterns of rivers, canals, and urban areas. The water is a deep blue, with some lighter blue patches indicating sediment or shallow areas. The sky is dark blue with some white clouds visible on the right side.

Fundamental of Remote Sensing

Tong Si Son

Email: tong-si.son@usth.edu.vn

Specification: Remote sensing and Applications

Structure Of The Course

- **Time**

35 hours: Lecturer: 20 h, Practice: 15

- **Contents**

1. Overview of Remote sensing
2. Physical basis of Remote sensing
- 3. Structure of remote sensing systems**
4. Satellite image characteristics
5. Satellite image pre-processing
6. Satellite image classification and accuracy assessment

Section 3. Structure of remote sensing systems

(Material by Prof. Gabriëlle De Lannoy, KU Leuven)

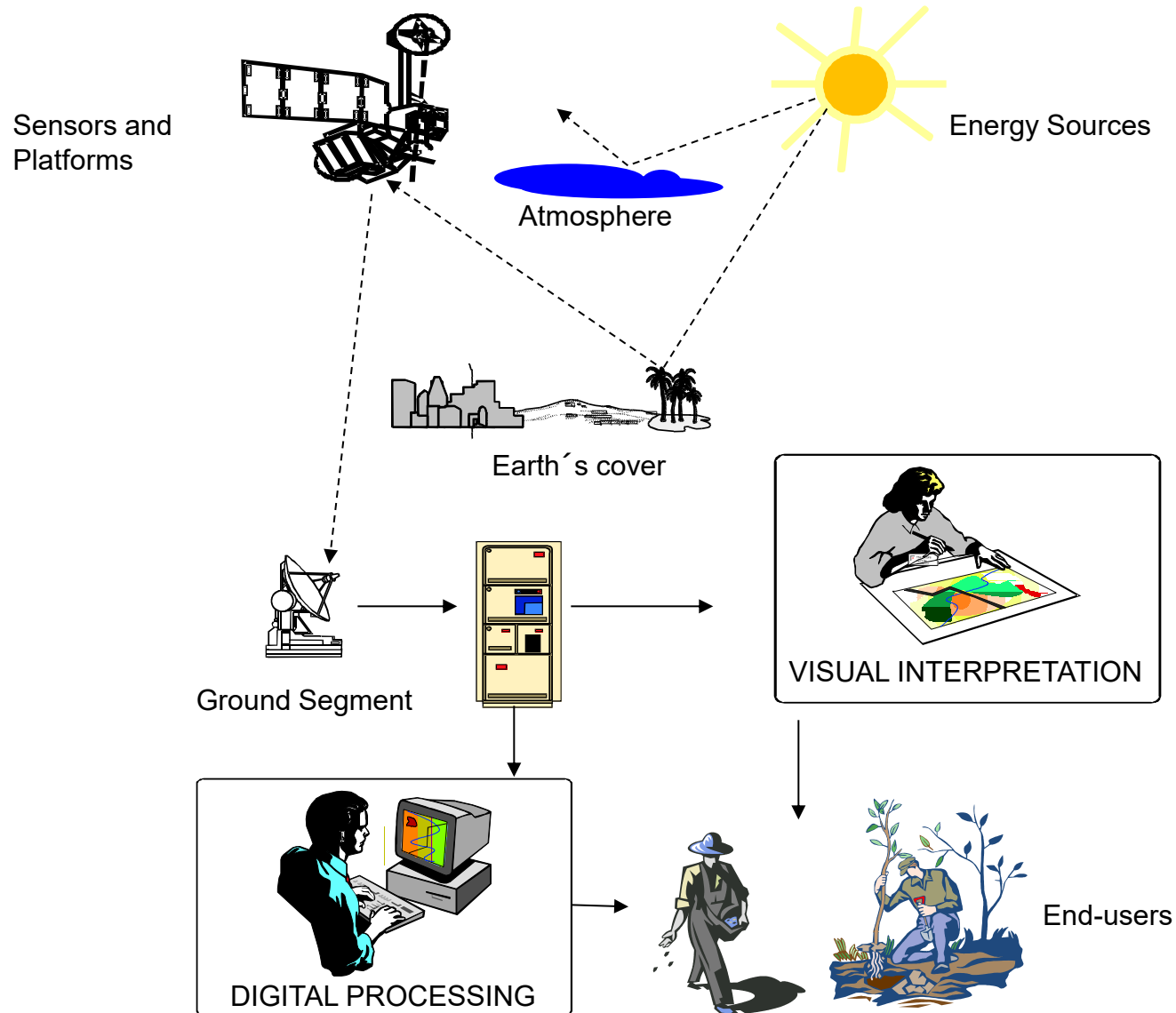
3.1 Components of a remote sensing system

3.2 Sensors

3.3 Platforms

3.4 Ground segment

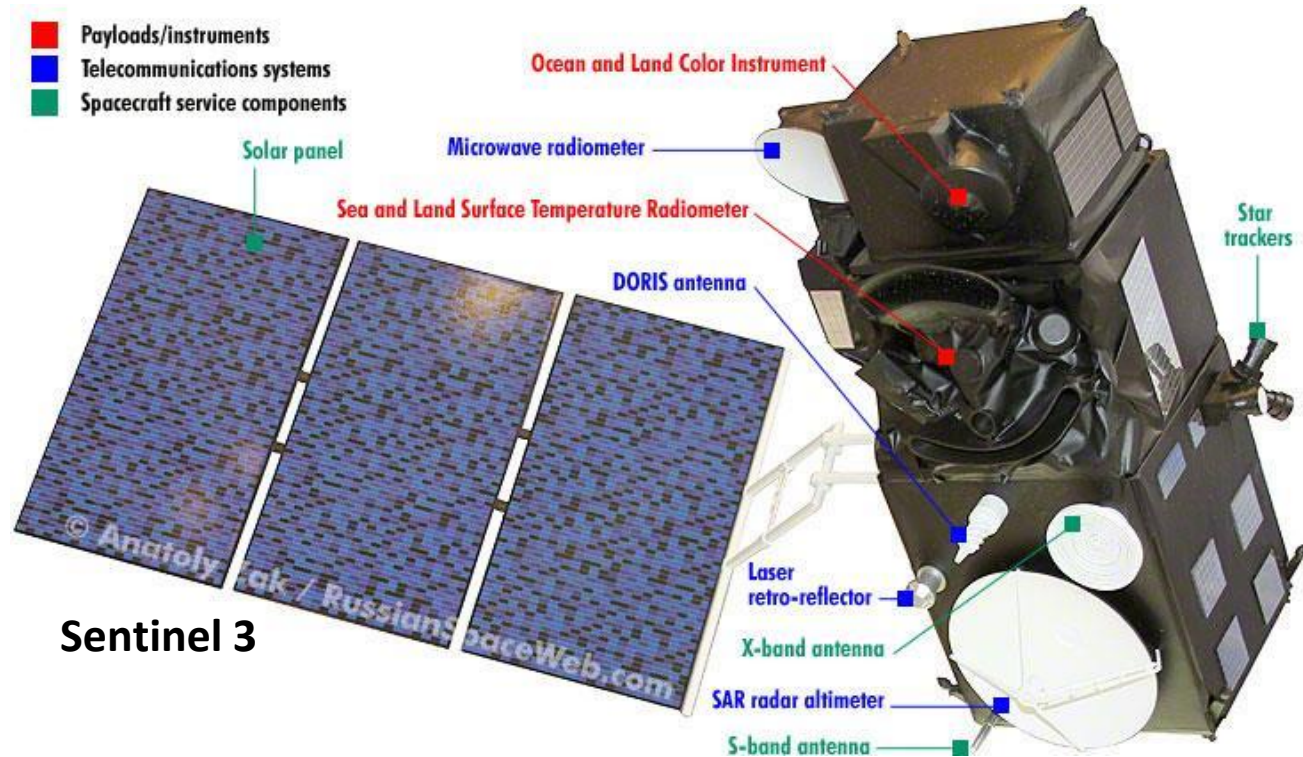
3.1 Components of a remote sensing system



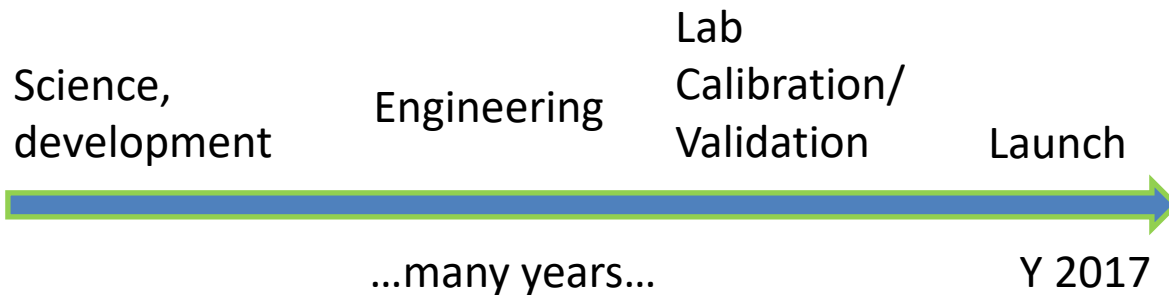
(Chuvieco and Huete, 2009)

3.2 SENSORS: ACTIVE - PASSIVE

- Sensor + platform >> spatial, temporal, spectral, radiometric resolution

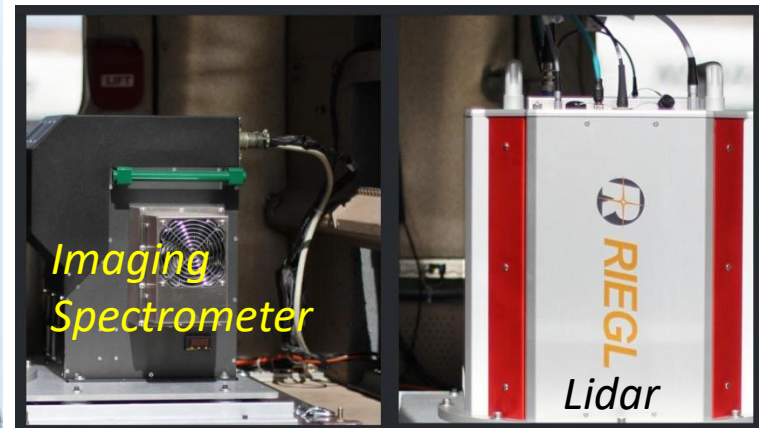
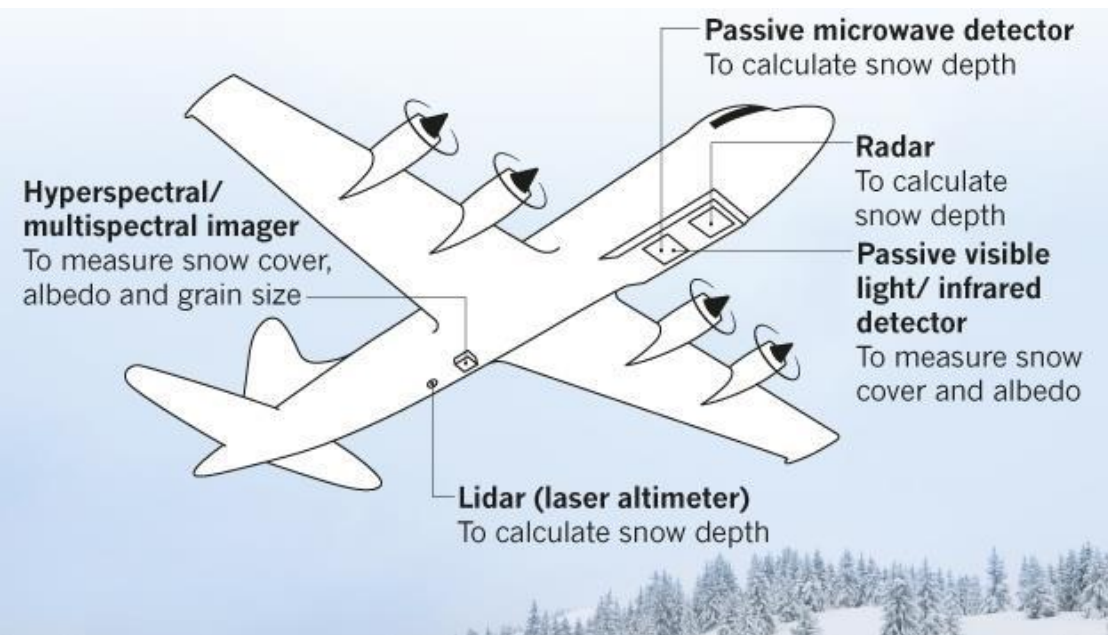


Sentinel 3



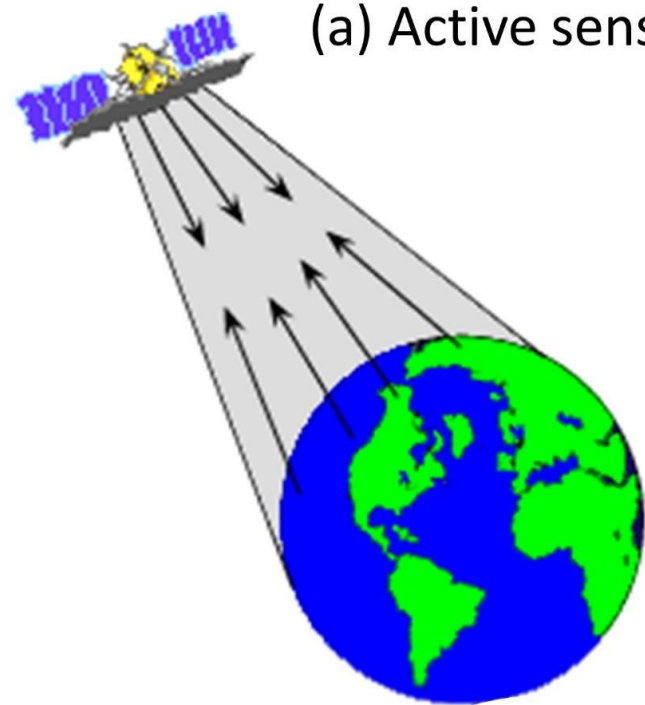
Sensors and Platforms

NASA/JPL's Airborne Snow Observatory (ASO)

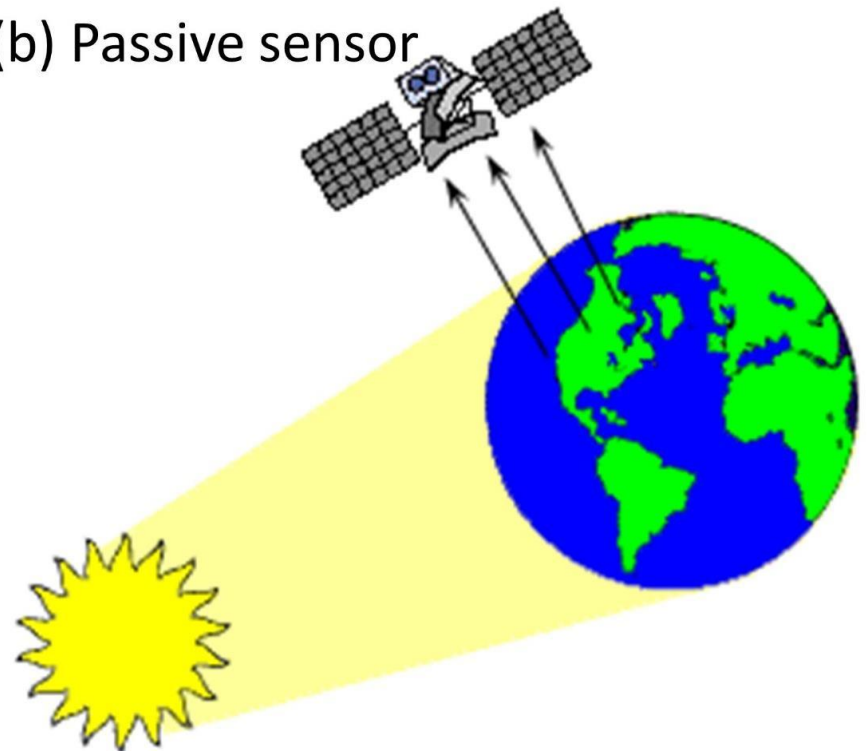


REMOTESENSING SENSORS

(a) Active sensor

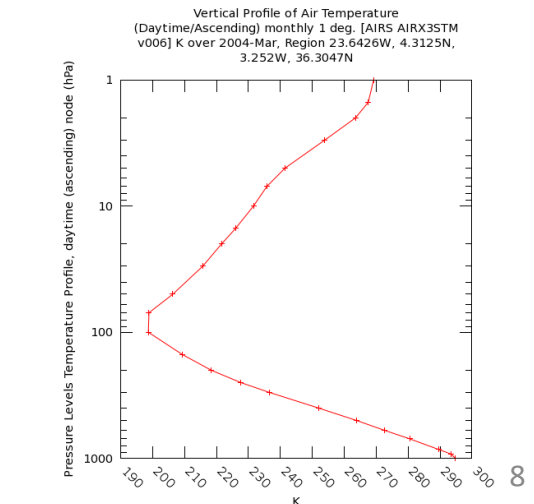
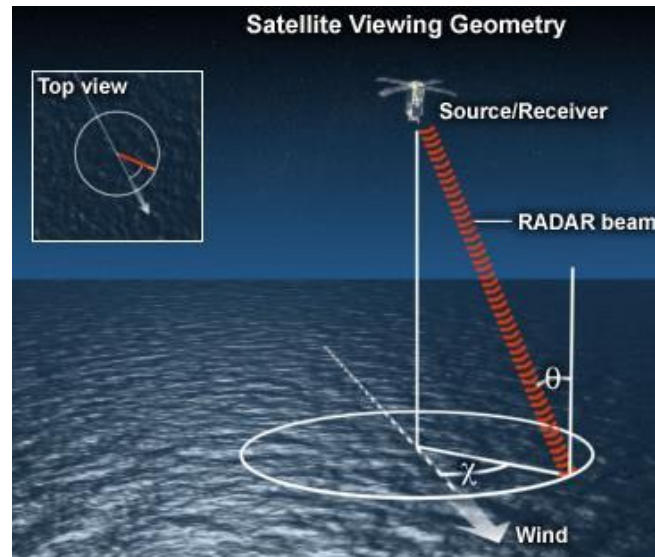


(b) Passive sensor



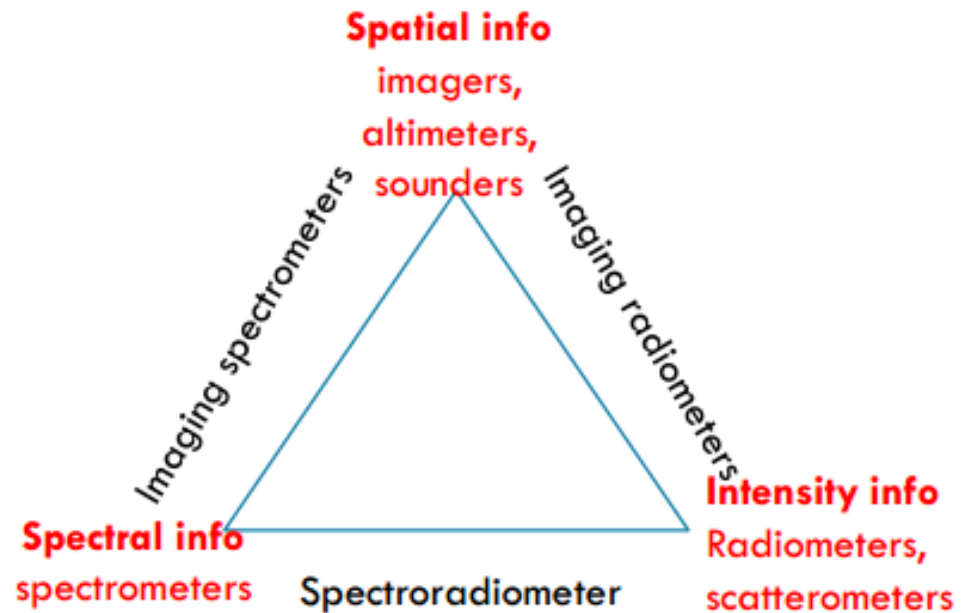
ACTIVE SENSORS

- Radar
- Ranging instrument
- Lidar
- Scatterometer
- Laser altimeter
- Sounder

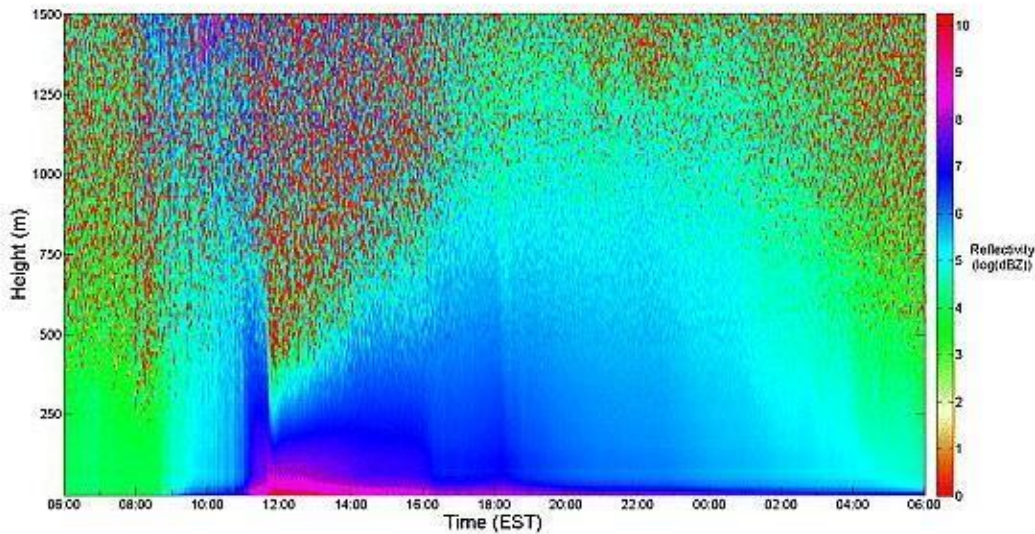


ACTIVE SENSORS

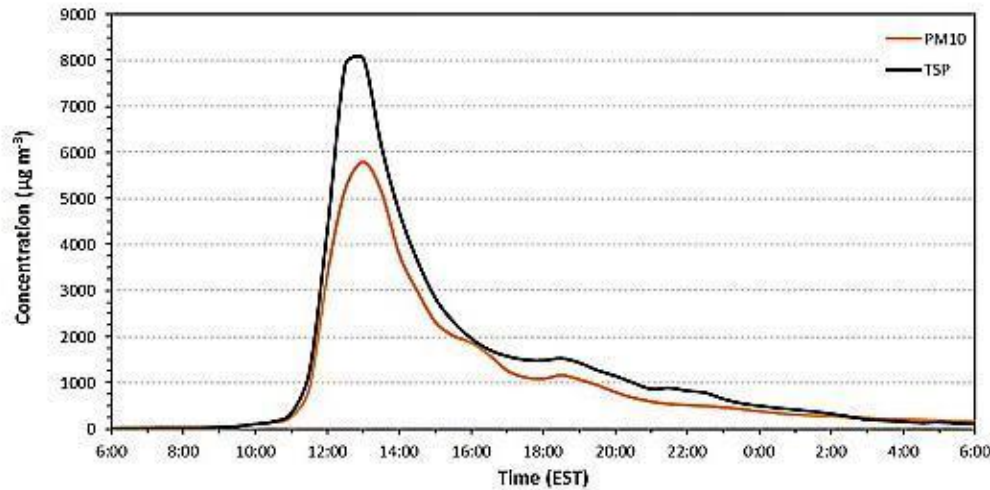
- Radiometer
 - Imaging radiometer
 - Hyperspectral radiometer
 - Spectroradiometer
- Spectrometer



NON-IMAGING SENSORS



Laser ceilometer for dust profiling

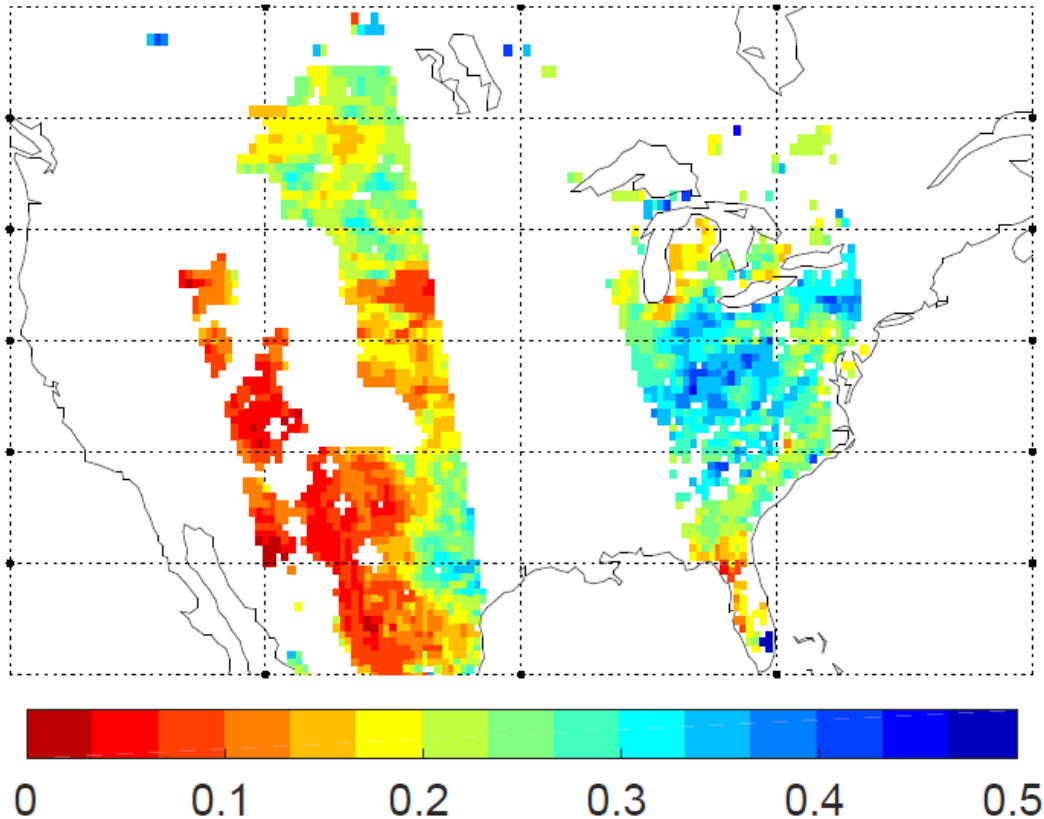


1
footprint



- Fixed location, spatial resolution dependent on IFOV
- Multiple observations in time

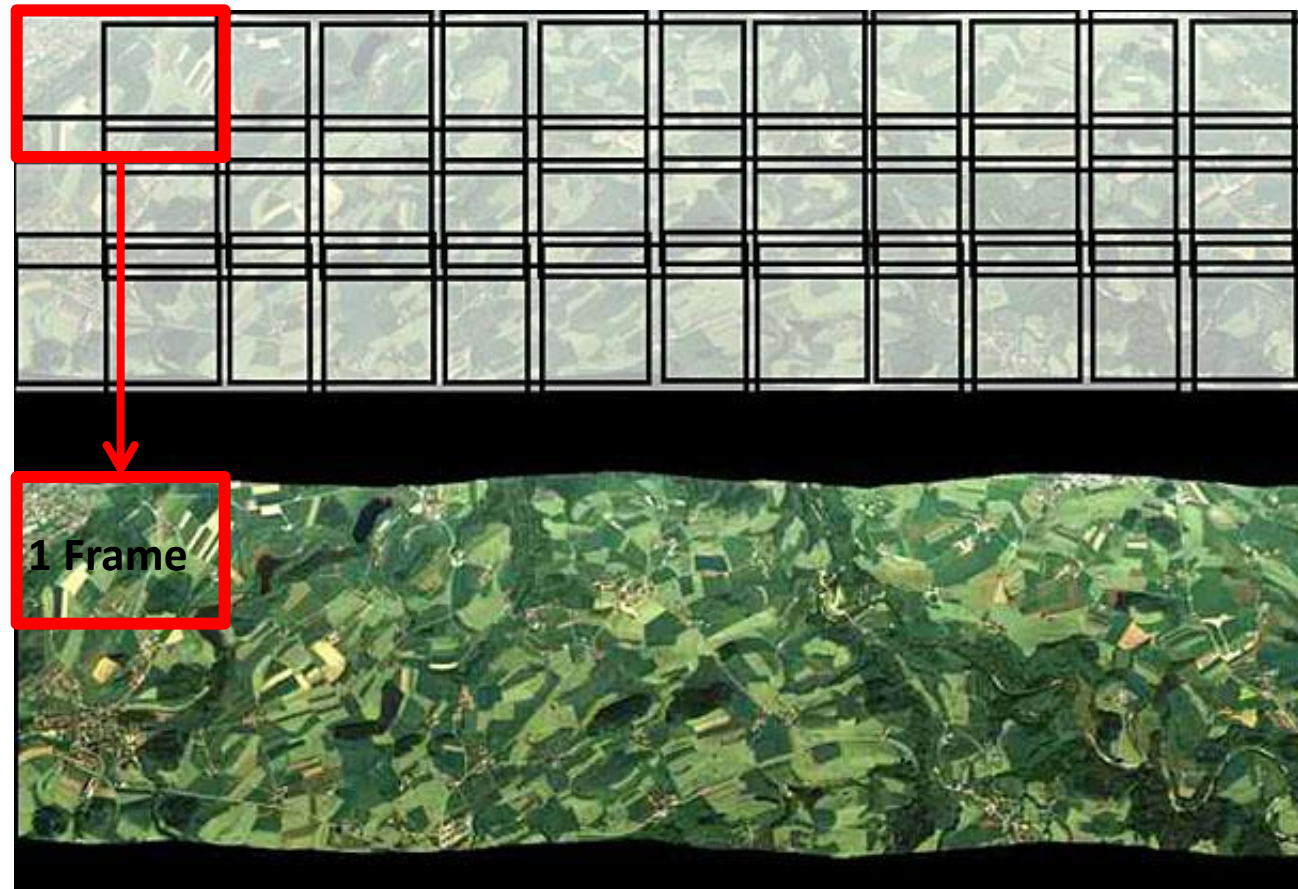
IMAGING SENSORS



- **Raster** w/ (mostly) digital data, sampled and quantized representation of target scene
- Digital number = radiance in that pixel (derived from circular IFOV)

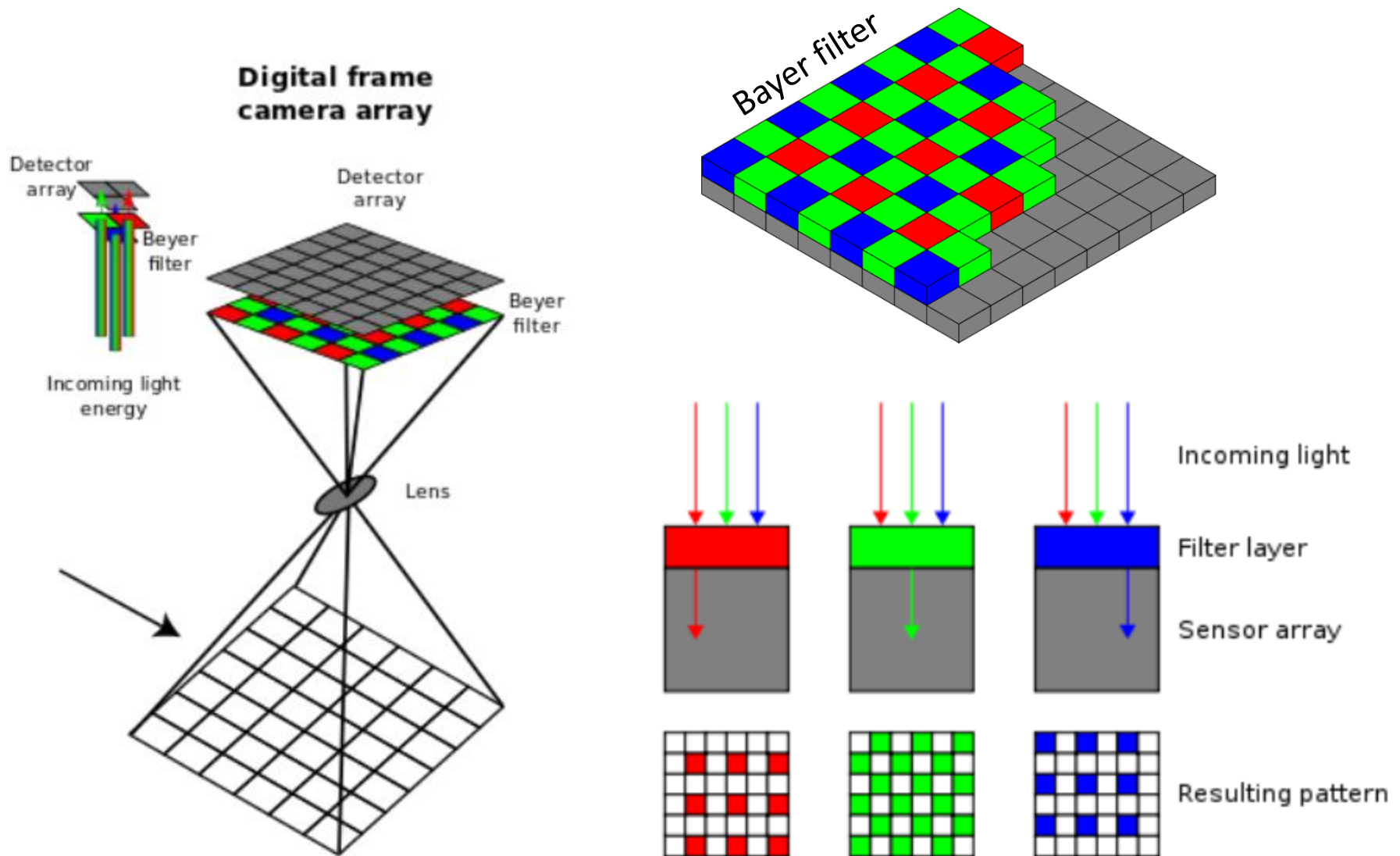
IMAGING SENSORS

- Framing cameras
- Across-track scanners
- Along-track scanners
- Imaging radiometers
- Imaging spectrometers



>>> Active/Passive

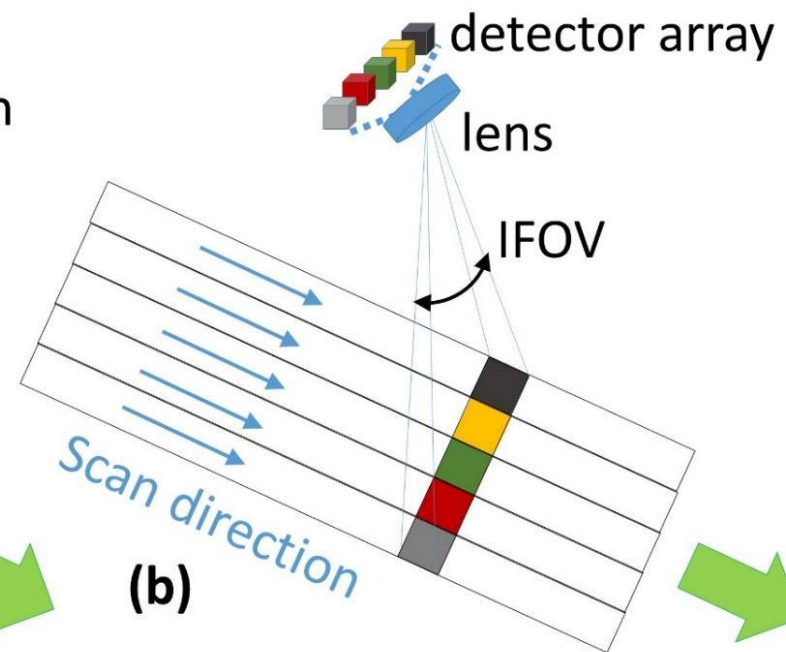
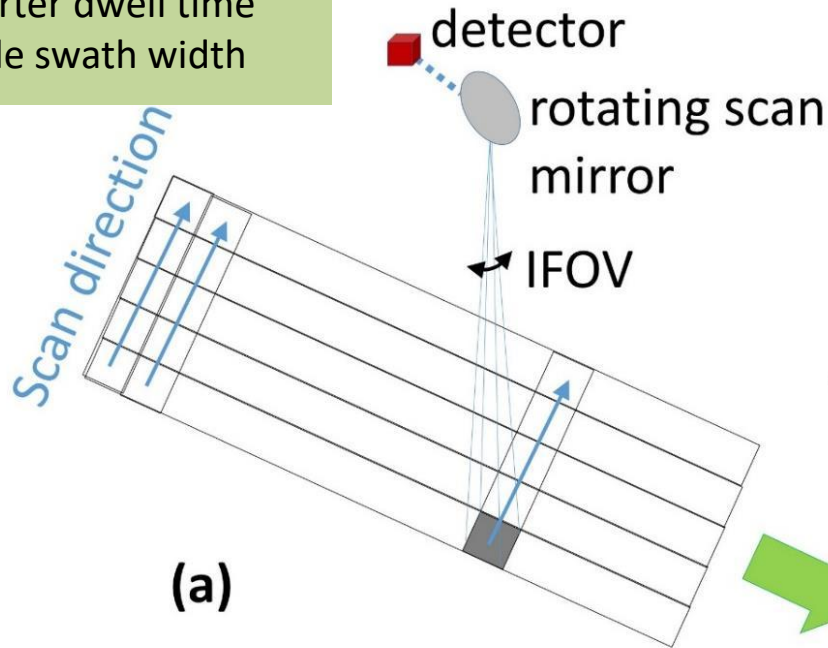
IMAGING SENSORS



IMAGING SENSORS

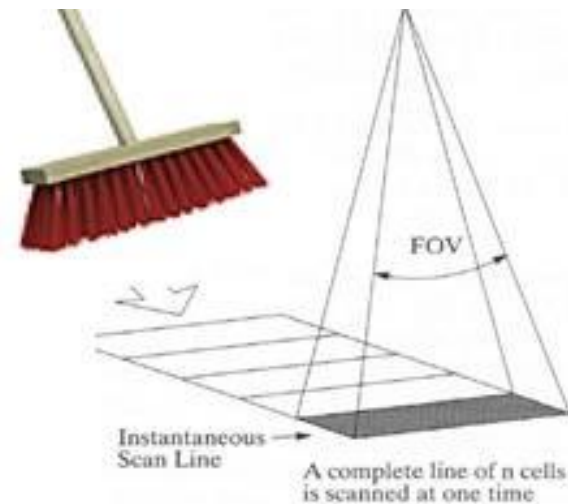
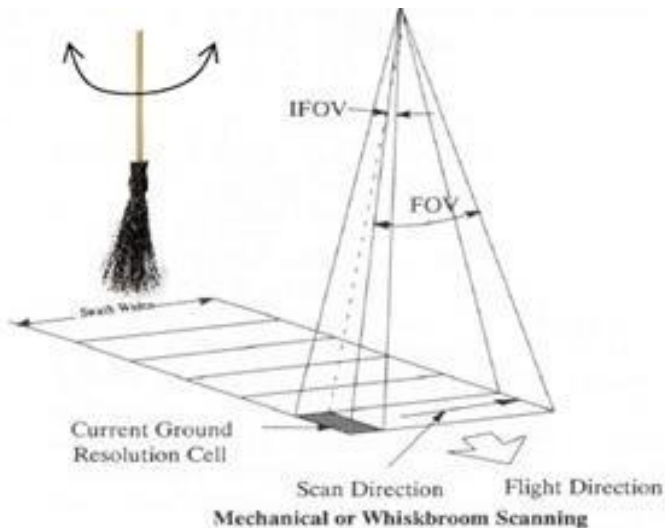
Across-track Scanner

- Simple optical system
- Shorter dwell time
- Wide swath width



Along-track Scanner

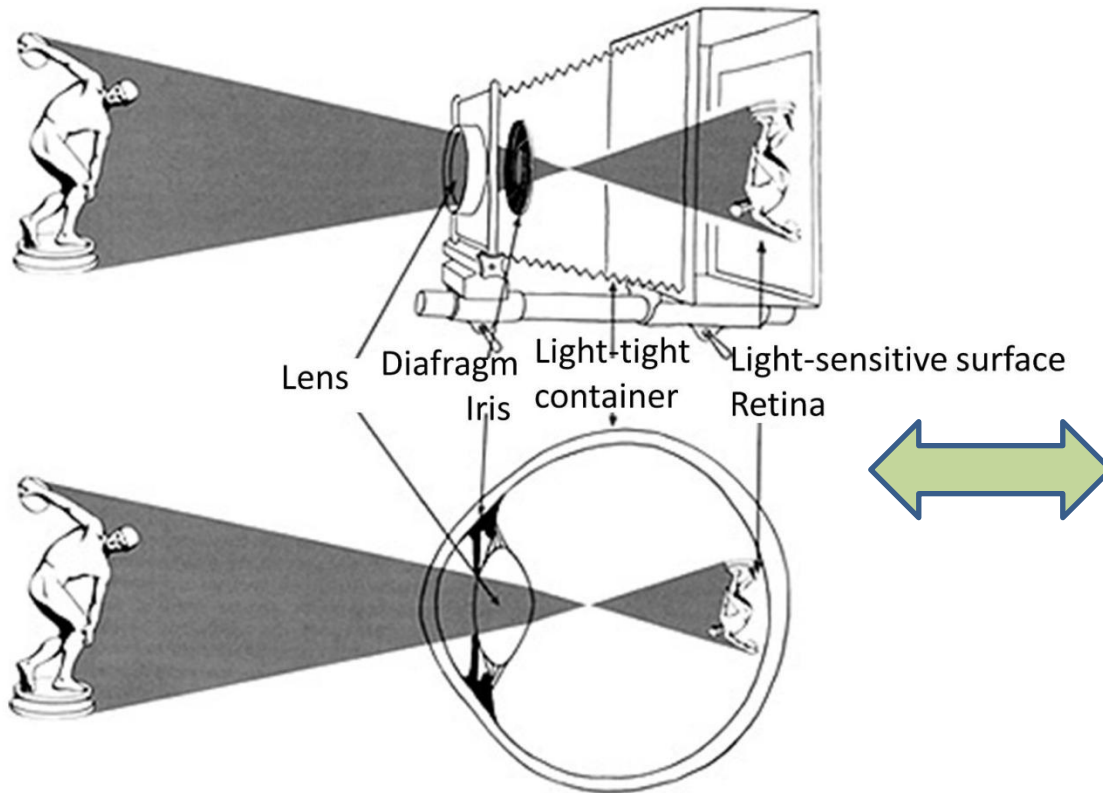
- Complex optical system (dispersion grating...)
- Longer dwell time
- Narrow swath width



SENSOR COMPONENTS

- Lenses, antennas: focus and collect EMR
- Imaging surface and antennas to capture EMR
- Filters, prisms, gratings, multiplexers: control wavelength
- A/D converter
- Housing/body to hold all components together

SENSOR COMPONENTS

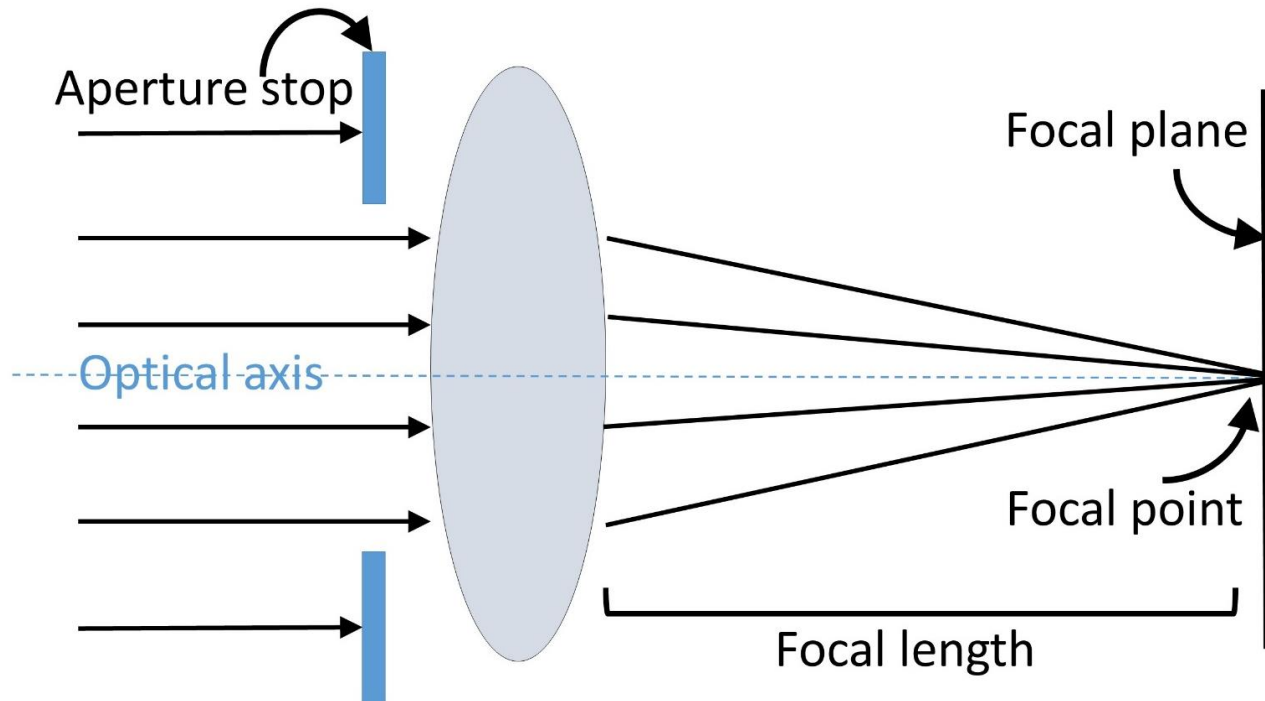


UV-VIS-NIR
Optical



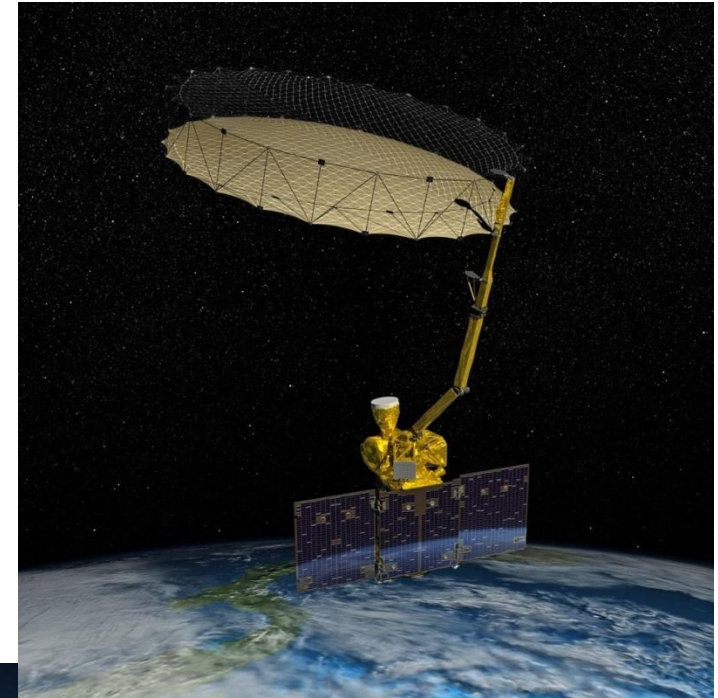
Microwaves

LENSES AND OPENINGS



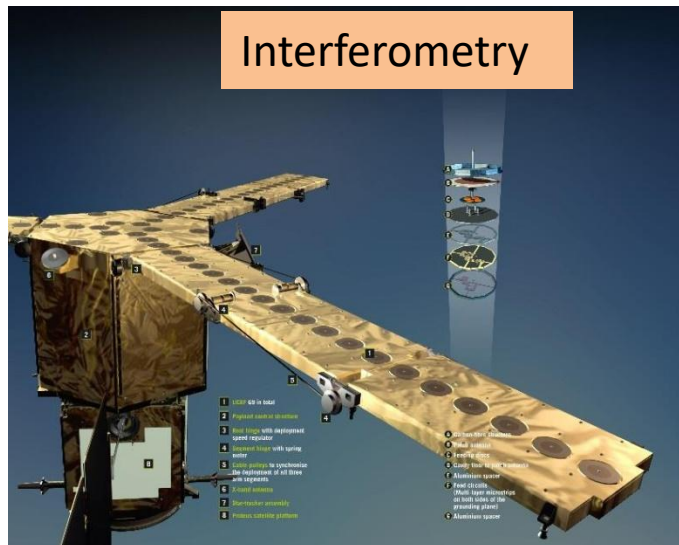
Diameter of lens
>>> diffraction
>>> lower limit to
Spatial resolution

ANTENNAS

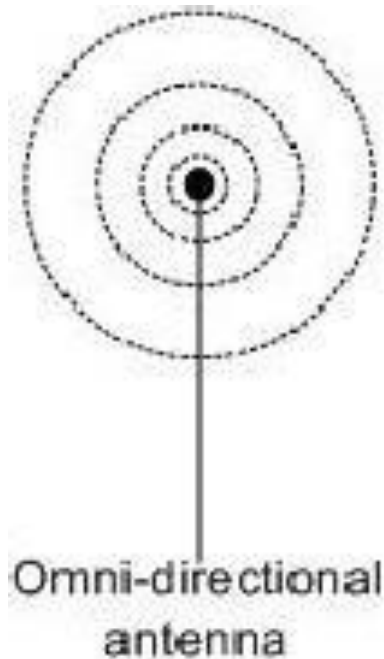


Converts electric power into radiowaves or converts radiowaves into electric power

transmitter/receiver



ANTENNAS



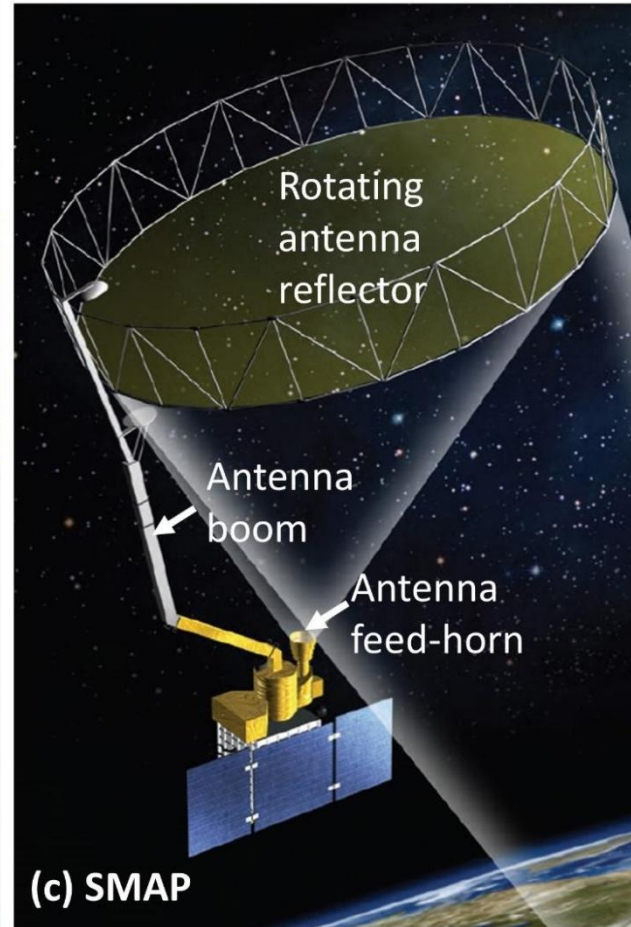
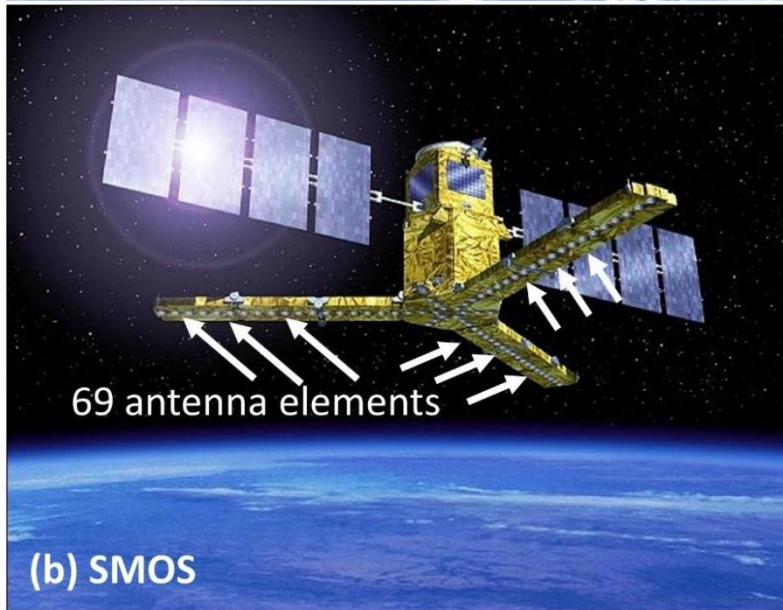
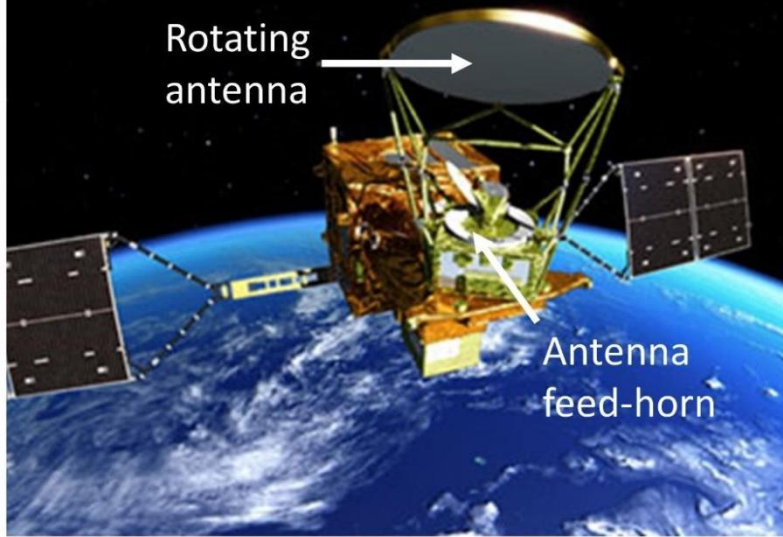
- Omnidirectional: transmit/receive power in all directions
- Directional: transmit/receive power preferentially in 1 direction



- Mostly used in RS
- Require parabolic reflectors or horns to direct radiowaves in a beam or other desired radiation pattern

ANTENNAS

(a) AMSR-2

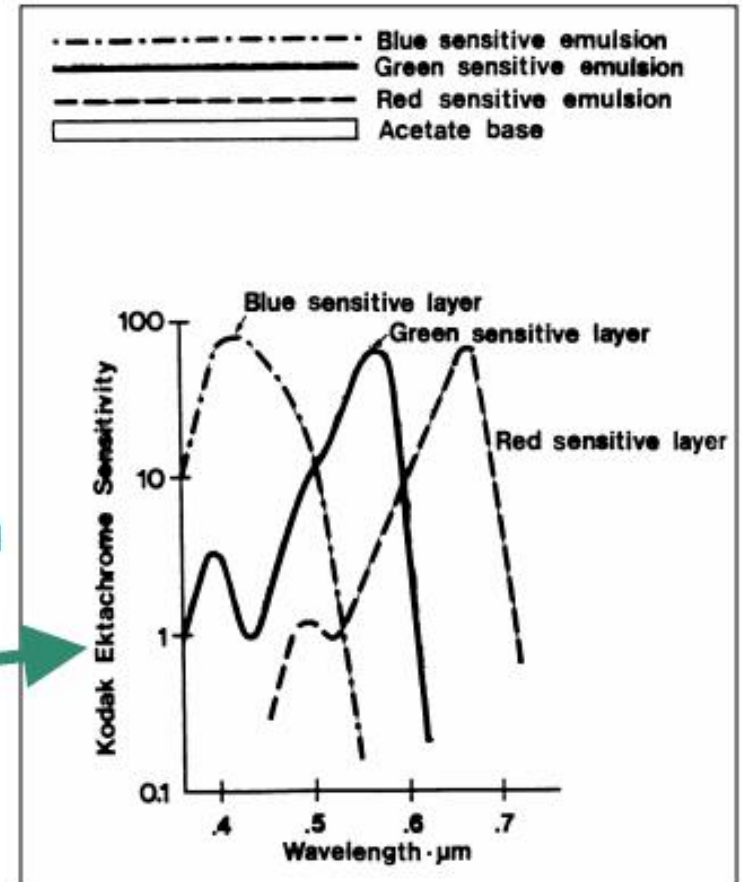
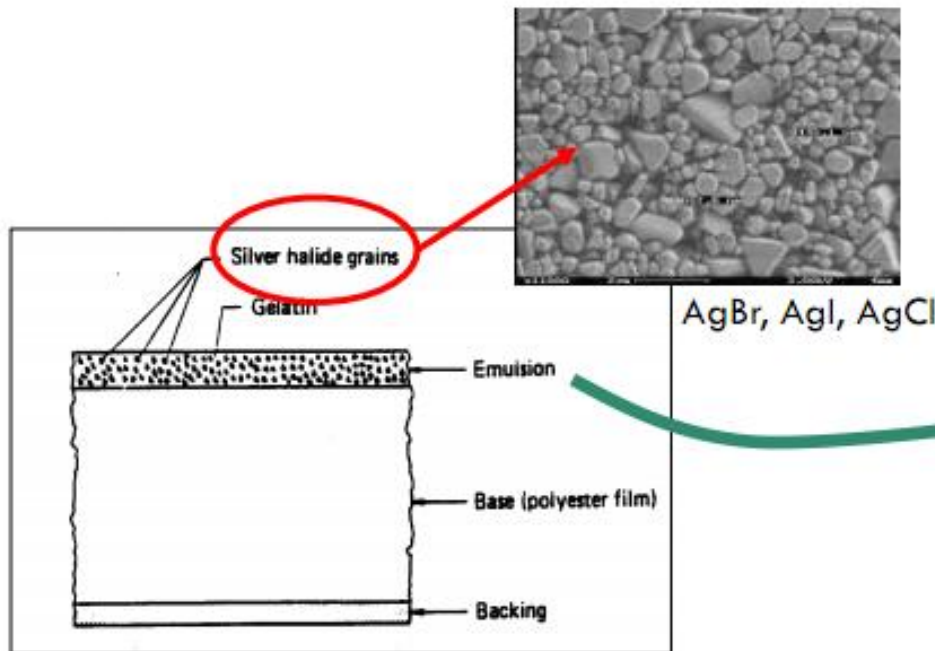


- **Diameter of antenna**
>>> diffraction
>>> lower limit to spatial resolution

IMAGING SURFACE, DETECTORS

Analogue (film) vs. digital (integrated circuits)

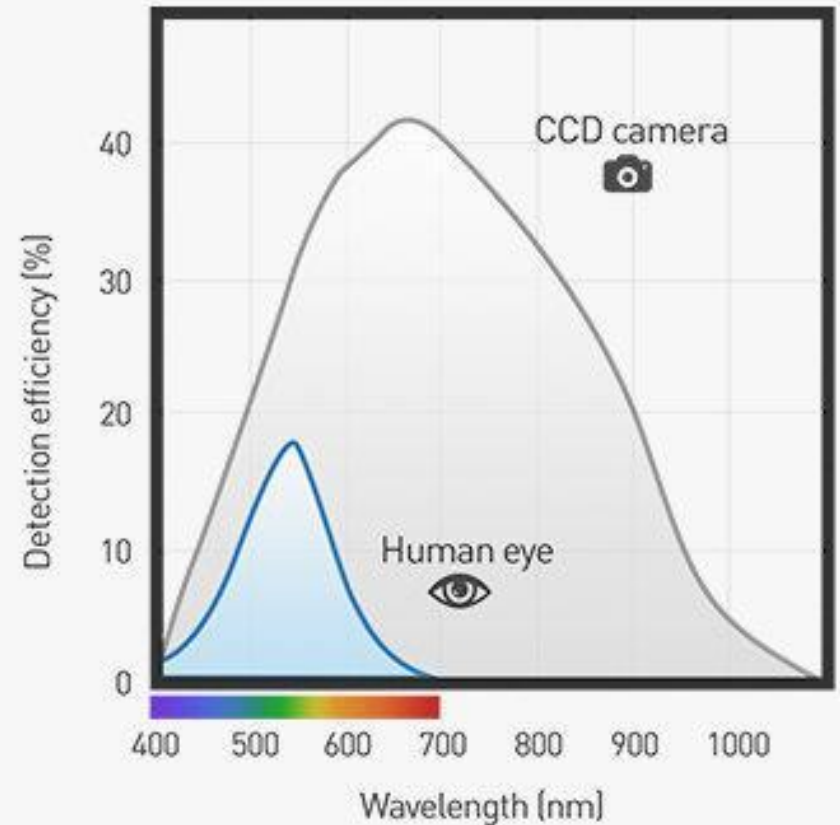
Film >> photographic process



IMAGING SURFACE, DETECTORS

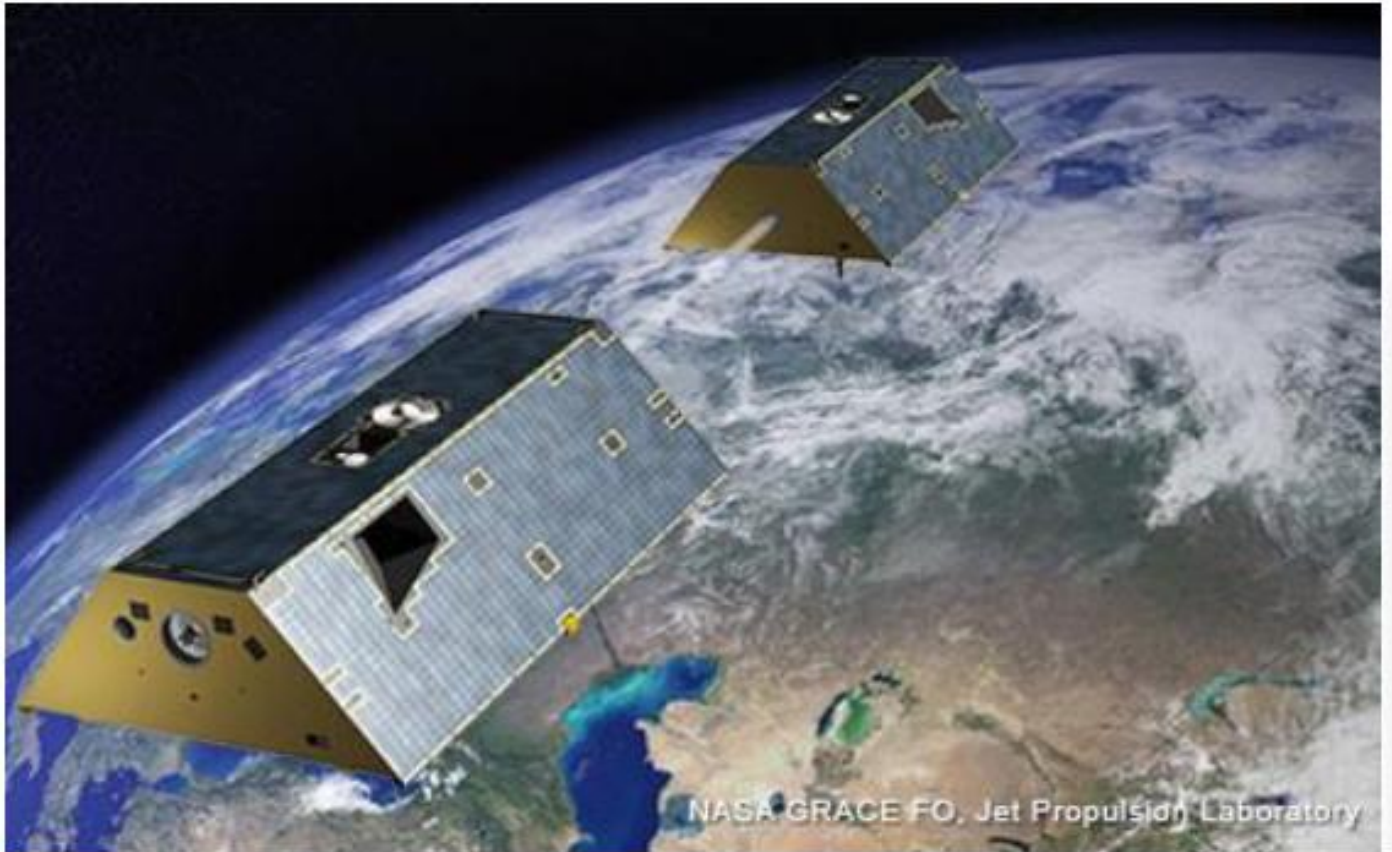
Digital >>> electronics

- Charged coupled device (CCD)
- Complementary metal oxide semiconductor (CMOS)
- Quantum well infrared photodetectors (QWIP)

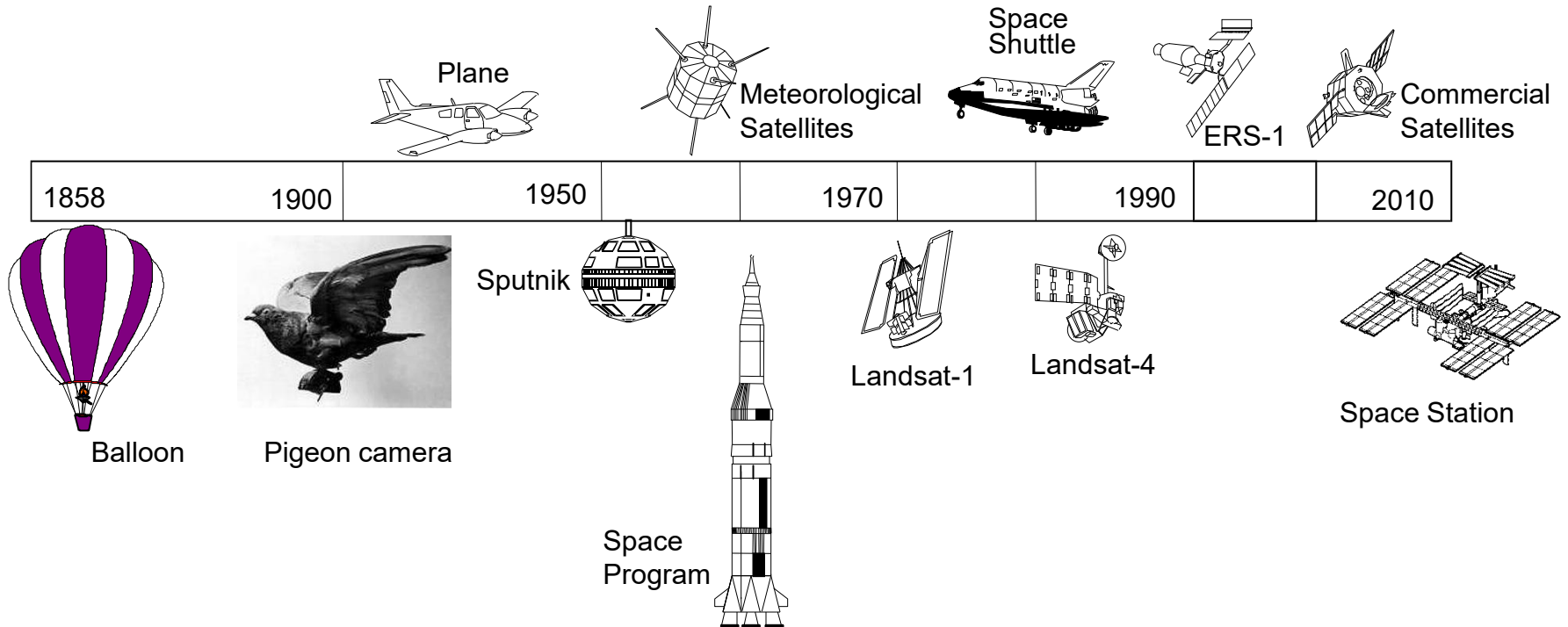


NON-EMR SENSORS

- Ultrasound, sonar (acoustic)
- Accelerometer
- Gravimeter

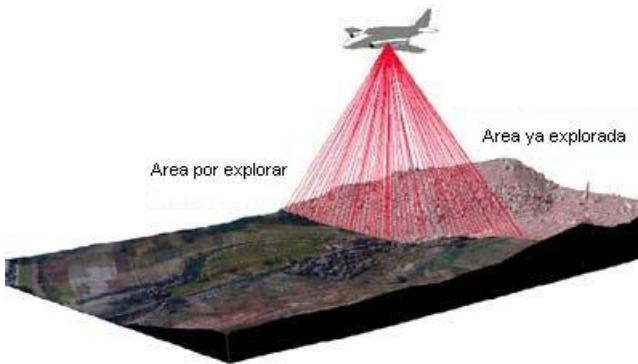


3.3 Platforms



Platforms

- Airborne (lidar, hyperspectral).
- Unmanned (UAV).
- Ground sensors



Platforms

- **Platform:**

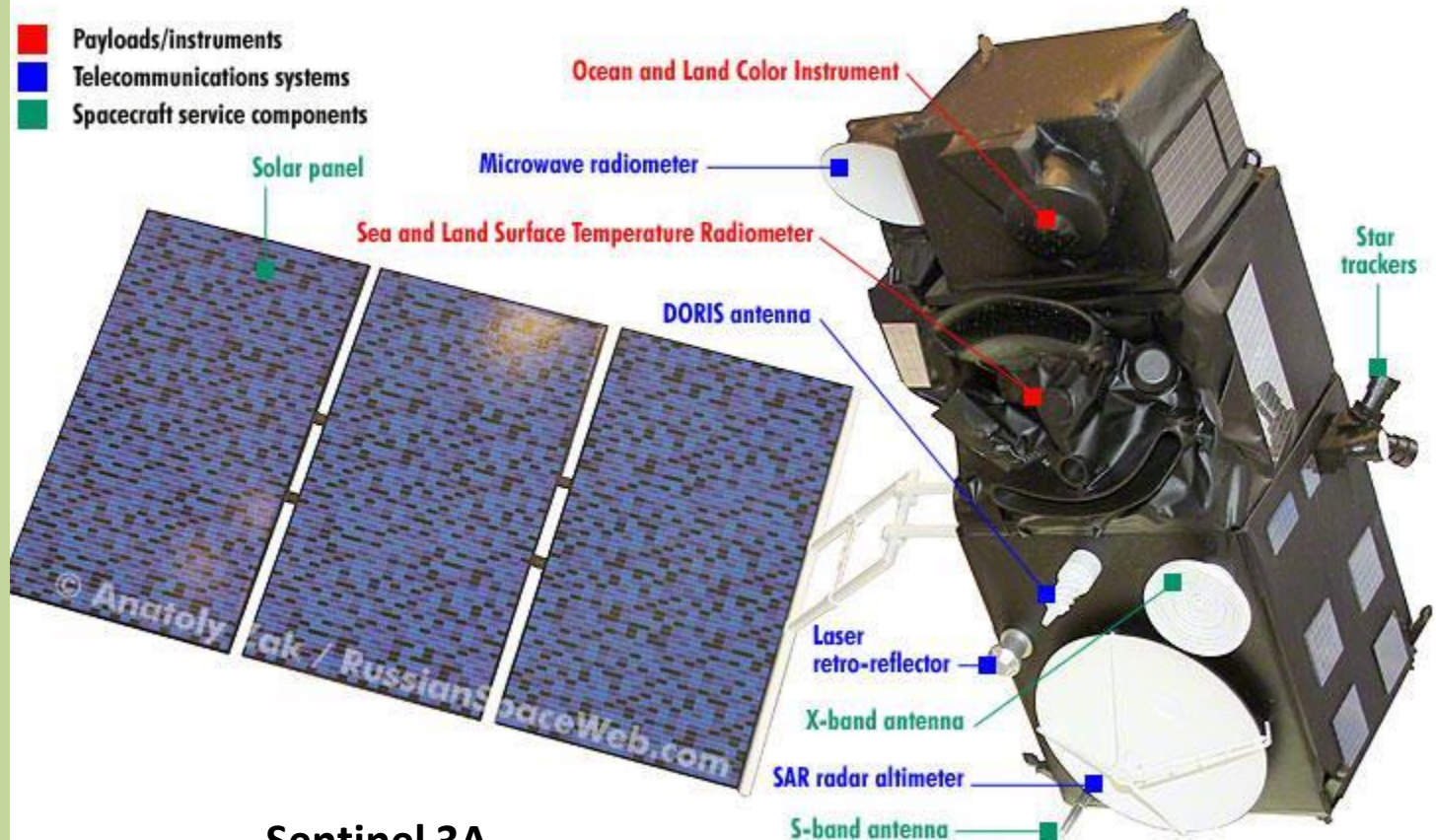
- Sensor

- Instruments to locate the sensor

- Storage capacity

- Communication

- Power



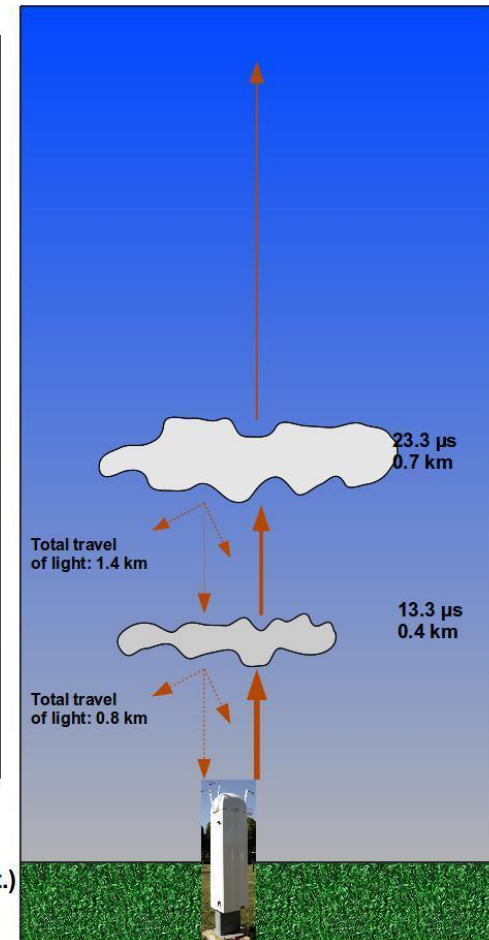
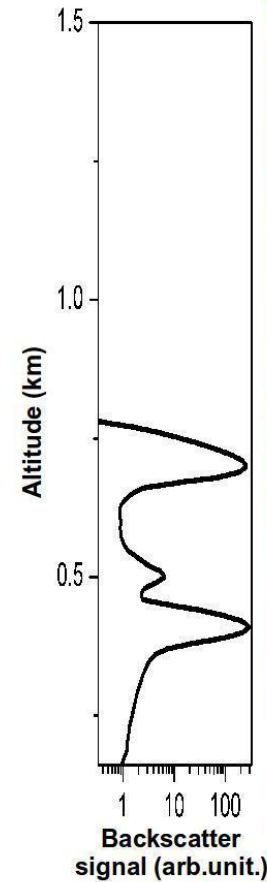
Sentinel 3A

Platforms

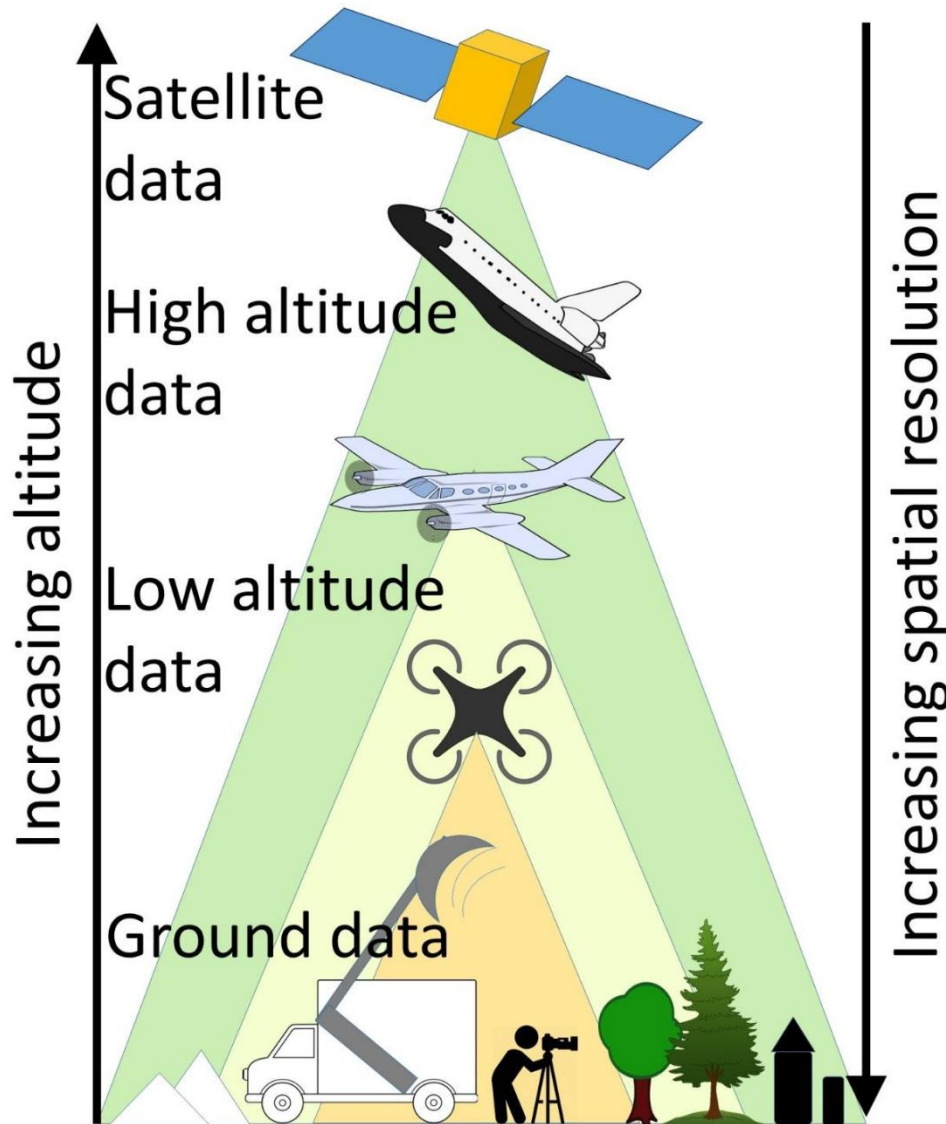
- **Position:**
Moving platforms



Static
platforms



Platforms



Advantages/
disadvantages
of different
platforms?

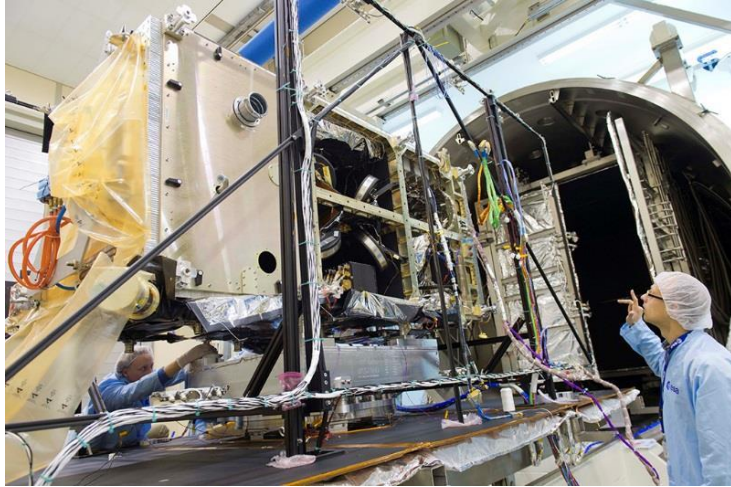
- Flexibility, Mobility
- Frequency, revisit
- Time series
- Cost, data availability
- Calibration, validation



Platforms

(Calibration)

Calibration:
(1) Lab calibration,
vacuum tests, ...



GOES-S

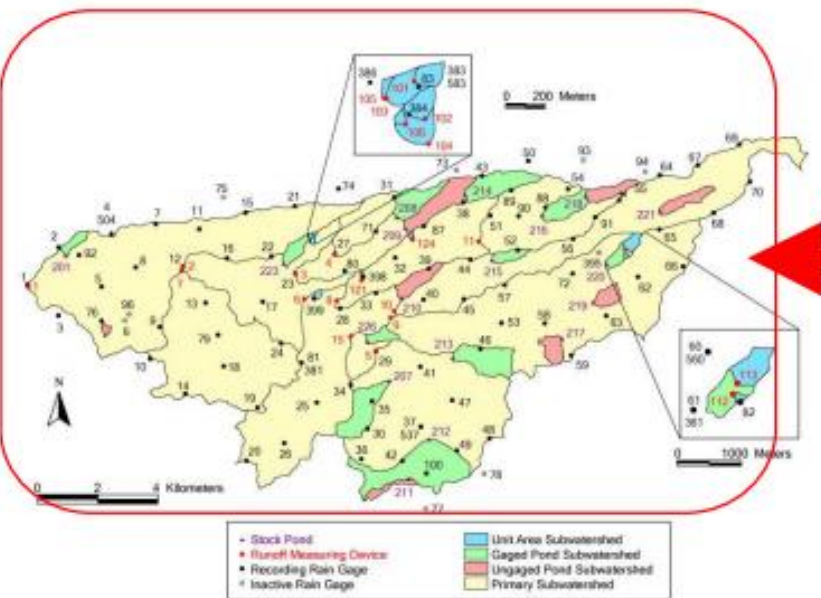


GOES-S

Platforms

(Calibration)

Calibration:
(2) in-orbit checkout (IOC)
(e.g. launch + 3 months)



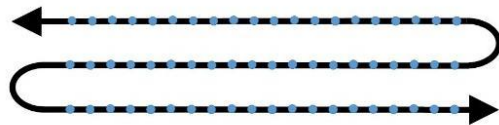
Walnut Gulch experimental watershed

Platforms

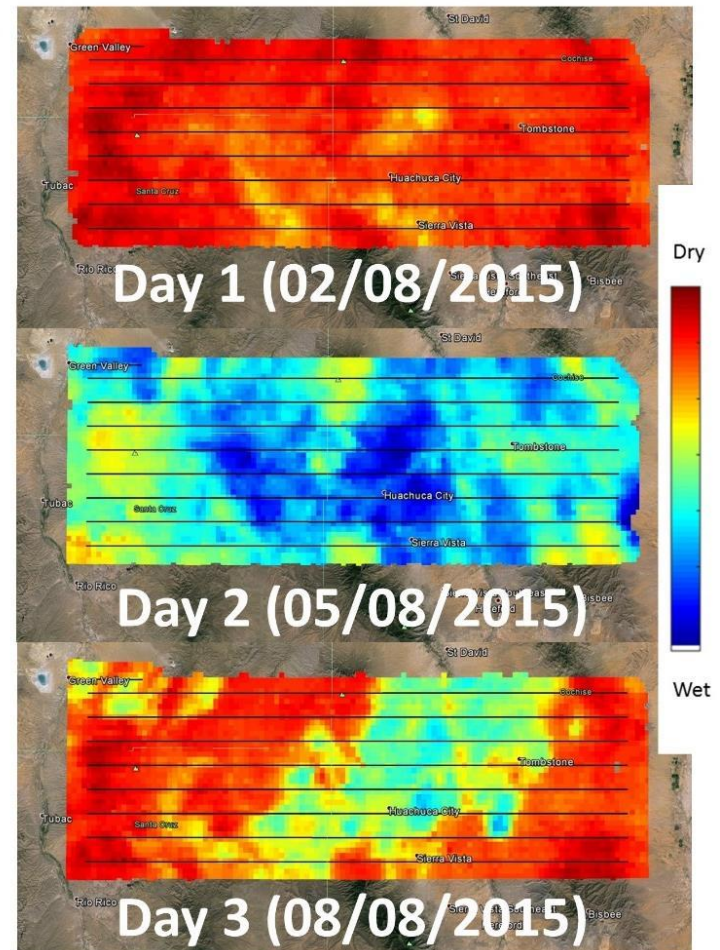
(Calibration)

Calibration:
(3) Ongoing during mission

(a)



(b)



SPACEBORNE - SATELLITES

4000+

Satellites
orbiting
the earth



1459

Active
satellites



52 %
Global
communication



11 %
Technology
demonstration



26 %
Earth
observation



7 %
Global
navigation



4 %
Space
science

OPERATION OF EO SYSTEMS



57 %
Government



6 %
Civilian



26 %
Military

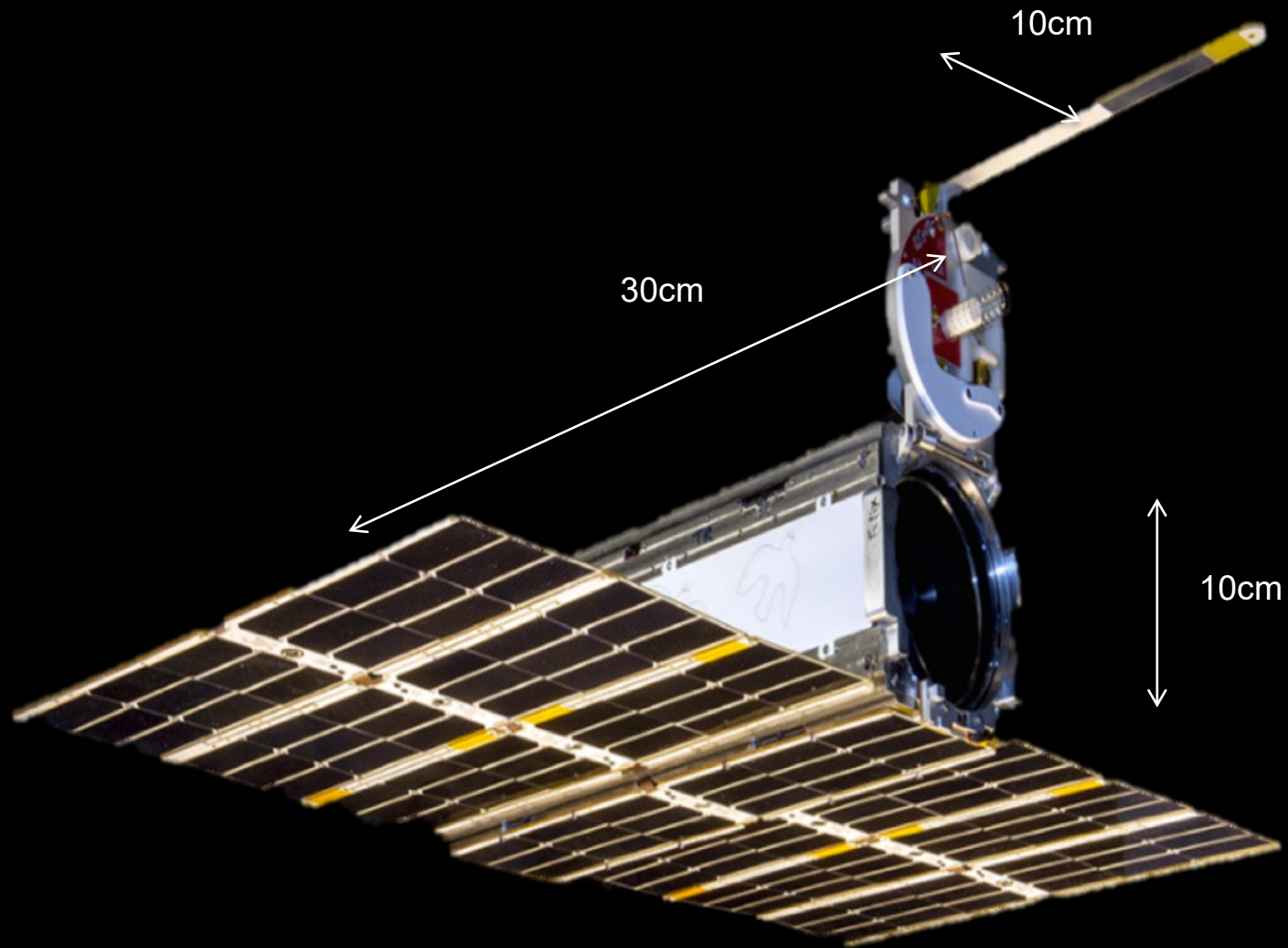


6 %
Commercial

Planet platform



+ THE DOVE SPACECRAFT



DOVE SATELLITE

3U CUBESAT

DIMENSIONS: 10 x 10 x 30 cm

WEIGHT: 4-5 kg

CAMERA CAPTURES

RED-GREEN-BLUE (RGB) AND
NEAR-INFRARED (NIR) IMAGERY

SPECTRAL BANDS:

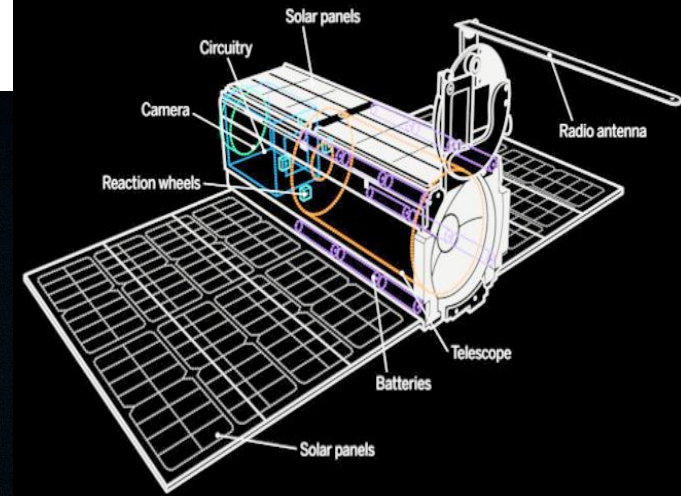
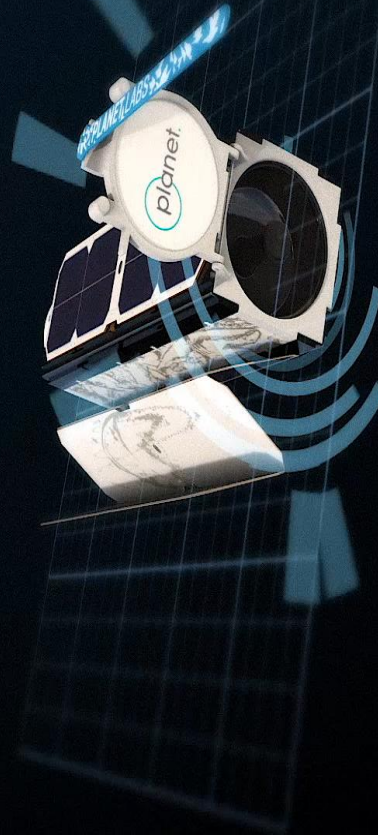
BLUE 455 – 515*

GREEN 500 – 590*

RED 590 – 670*

NIR (NEAR INFRARED) 780 – 860

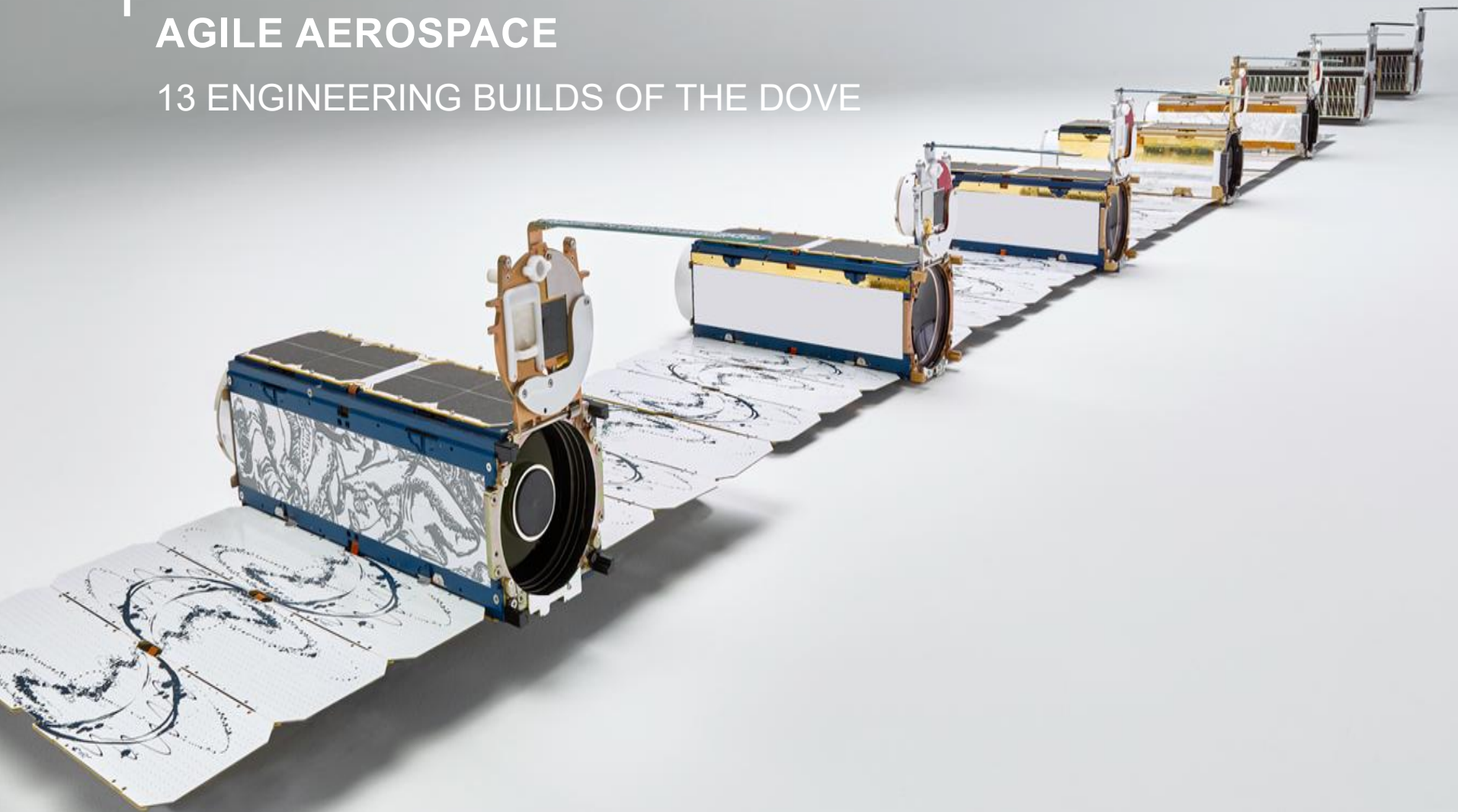
* λ_{fwhm}





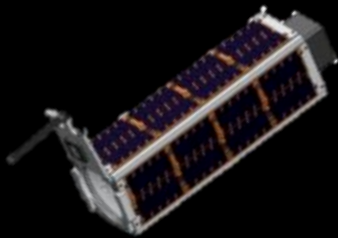
AGILE AEROSPACE

13 ENGINEERING BUILDS OF THE DOVE

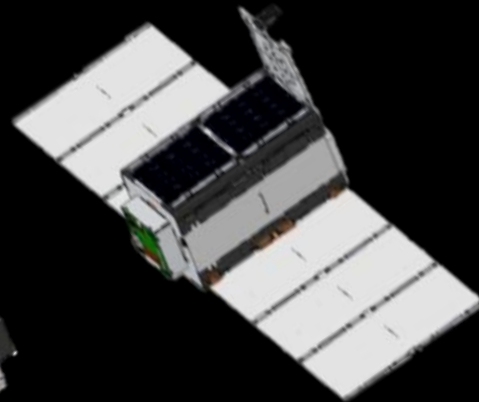


+ AGILE AEROSPACE

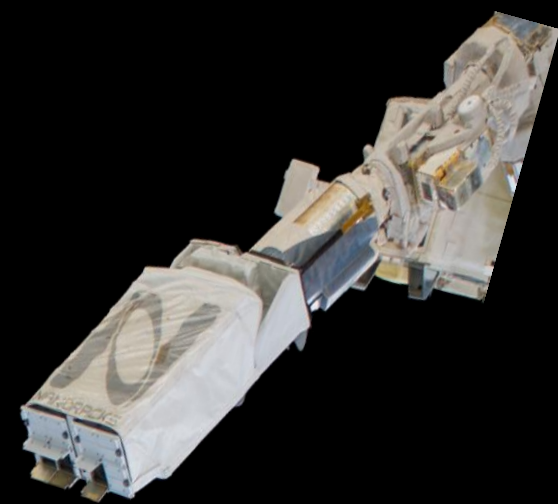
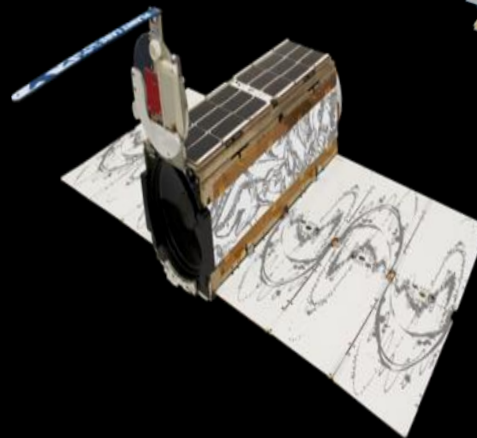
BUILD 1
APR 2012



BUILD 6
APR 2013



BUILD 13
JUN 2015



13 Builds in only 3 years
Largest constellation of Earth-imaging satellites





20 DOVES PER WEEK

MANUFACTURING FACILITY



Video about the Planet satellites

D:\Electrical Books\SG (VietNam)_08May2018\vids

Counted to 2018:

14 SUCCESSFUL LAUNCHES

271 DOVES DEPLOYED



Platform

- USA: Landsat, GOES, Space shuttle, Terra, Aqua, Ikonos, Quickbird, Geoeye..
- ESA: ERS-1 y 2, Envisat, Meteosat, MSG, Sentinel..
- France: Spot-4, 5, 6, 7
- India: IRS-6, Insat, Cartosat.
- Canadá: Radarsat-1 y 2.
- Russia: Spin-2, Resurs, Meteor.
- Japan: ALOS, GMS, Aqua.
- Brazil - China: CBERS, Fen Yung.
- Korea, Israel, Indonesia, Argentina, Vietnam...

	Aqua	Landsat	GOES	OCO-2	Sentinel-1	SMAP
orbit	sun-synchronous, near-polar, 705 km, 1:30 pm ascending, 16 days	sun-synchronous, near-polar, 705 km, 10:10 am descending, 16 days	geostationary, 35,790 km	sun-synchronous, near-polar, 705 km, 13:15 pm ascending, 16 days	sun-synchronous, near-polar, 693 km, 12 days	sun-synchronous, near-polar, 685 km, 6:00 am descending, 8 days
instrument	AMSR-E (6.9 GHz, ..., 89 GHz), MODIS, AMSU-A, AIRS, HSB, CERES	OLI, TIRS	Imager, Sounder	OCO-2	C-SAR	Radiometer
swath width	1445 km (AMSR-E), 2330 km (MODIS), 1690 km (AMSU-A), 1650 km (AIRS, HSB), scan angle 78°(CERES)	190 km (OLI, TIRS)	20°W - 165°E (GOES-13 + GOES-15)	10.3 km	80 km (strip map), 250 km (interferometric wide swath), 400 km (extra-wide swath)	1000 km
revisit time	~ 2 days	~ 16 days	‘continuous’	~ 16 days	3 days at equator, < 1 day at high latitudes	2-3 days
ground pixel resolution	75-5 km (AMSR-E low-high frequency), 40 km (AMSU-A), 13.5 km (AIRS), 13 km (HSB), 20 km (CERES)	30 m (OLI), 100 m (TIRS)	1-10 km (Imager), 10-50 km (Sounder)	3 km	5 m (strip map), 5×20 m ² (interferometric wide swath), 25×100 m ² (extra-wide swath), 20×20 km ² , 5×20 m ² (wave mode)	40 km (3 dB)

AIRBORNE PLATFORMS: AIRCRAFTS

Temporal resolution more flexible than satellites

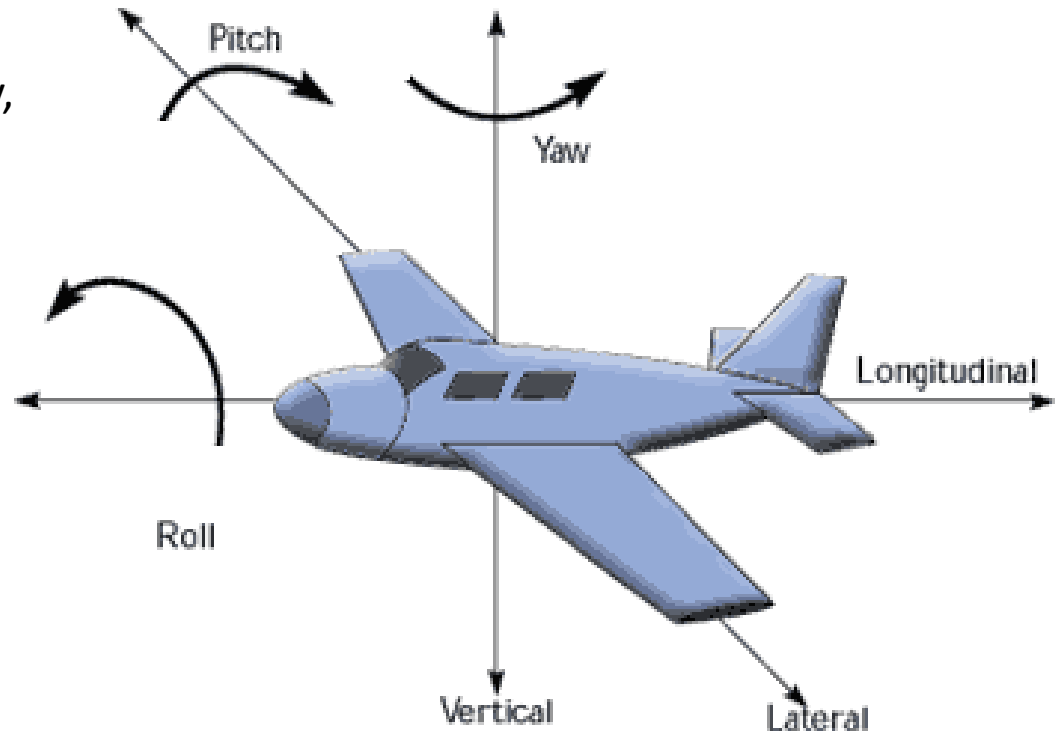
Limited by aircraft access, fuel capacity, atmospheric disturbances

Altitude, geometric position

>>Aerial photo processing

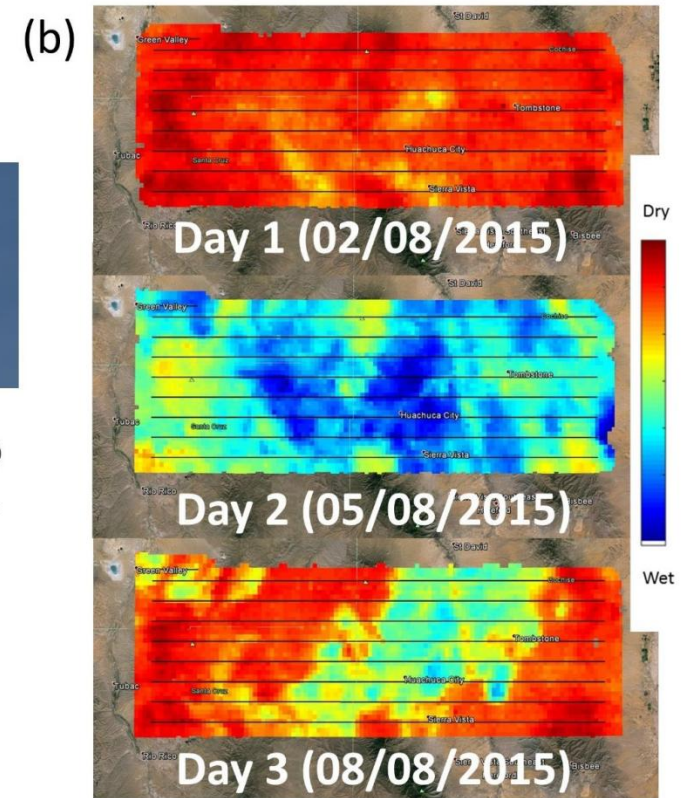
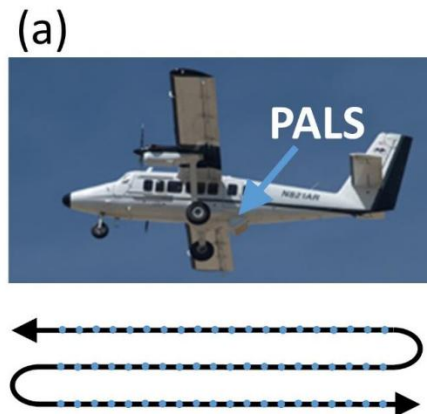
Key components

- Inertial Measurement Units
- Global Positioning System

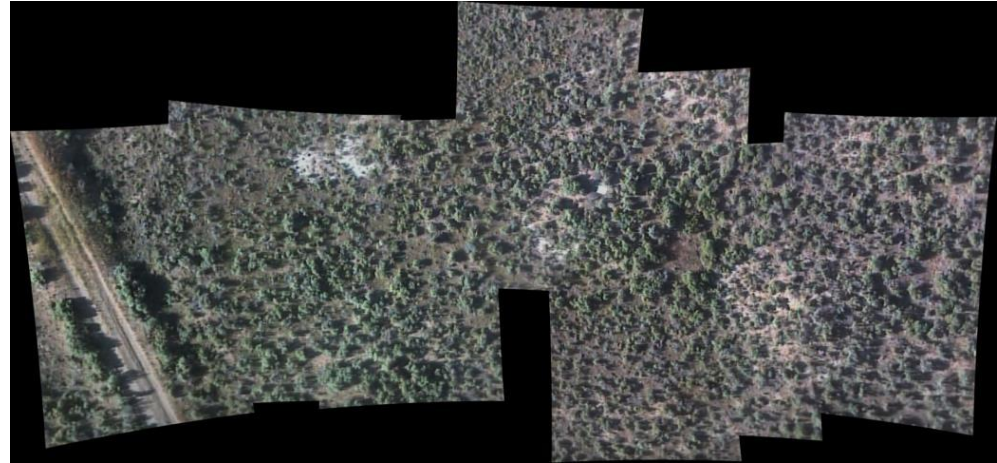


AIRBORNE PLATFORMS: AIRCRAFTS

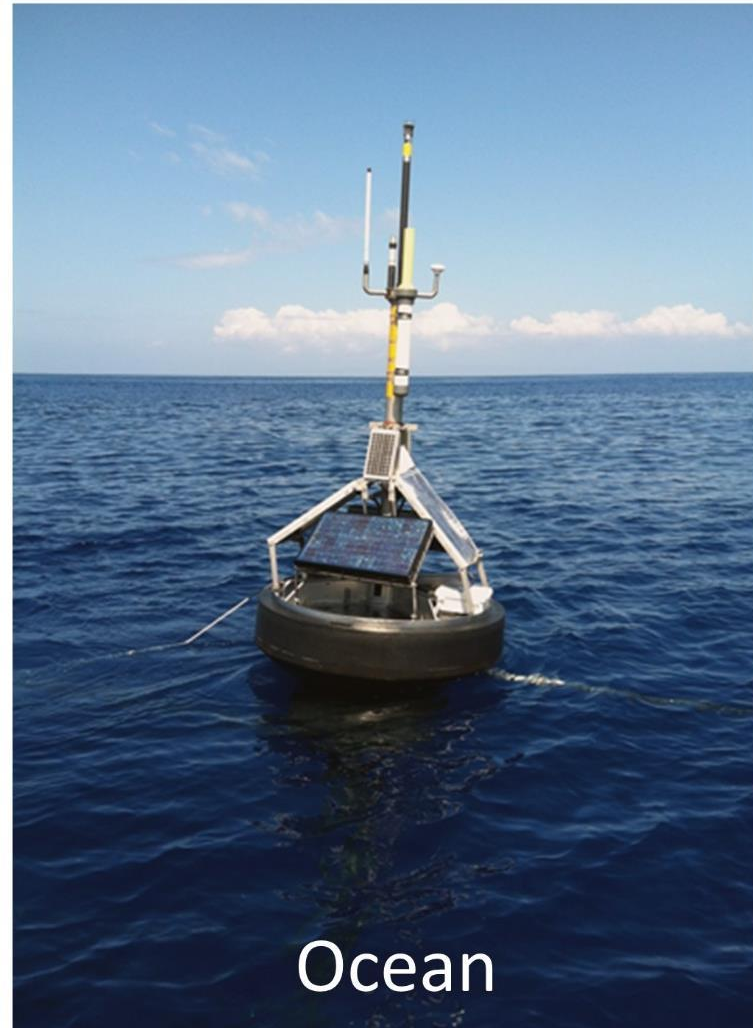
- Minimize costs
Minimize disturbances
Maximize data collection
- Planning flights:
Manually; GIS
- Data: overlap, stitching,
terrain/geometric
correction, stereoscopy



AIRBORNE PLATFORMS: UAS

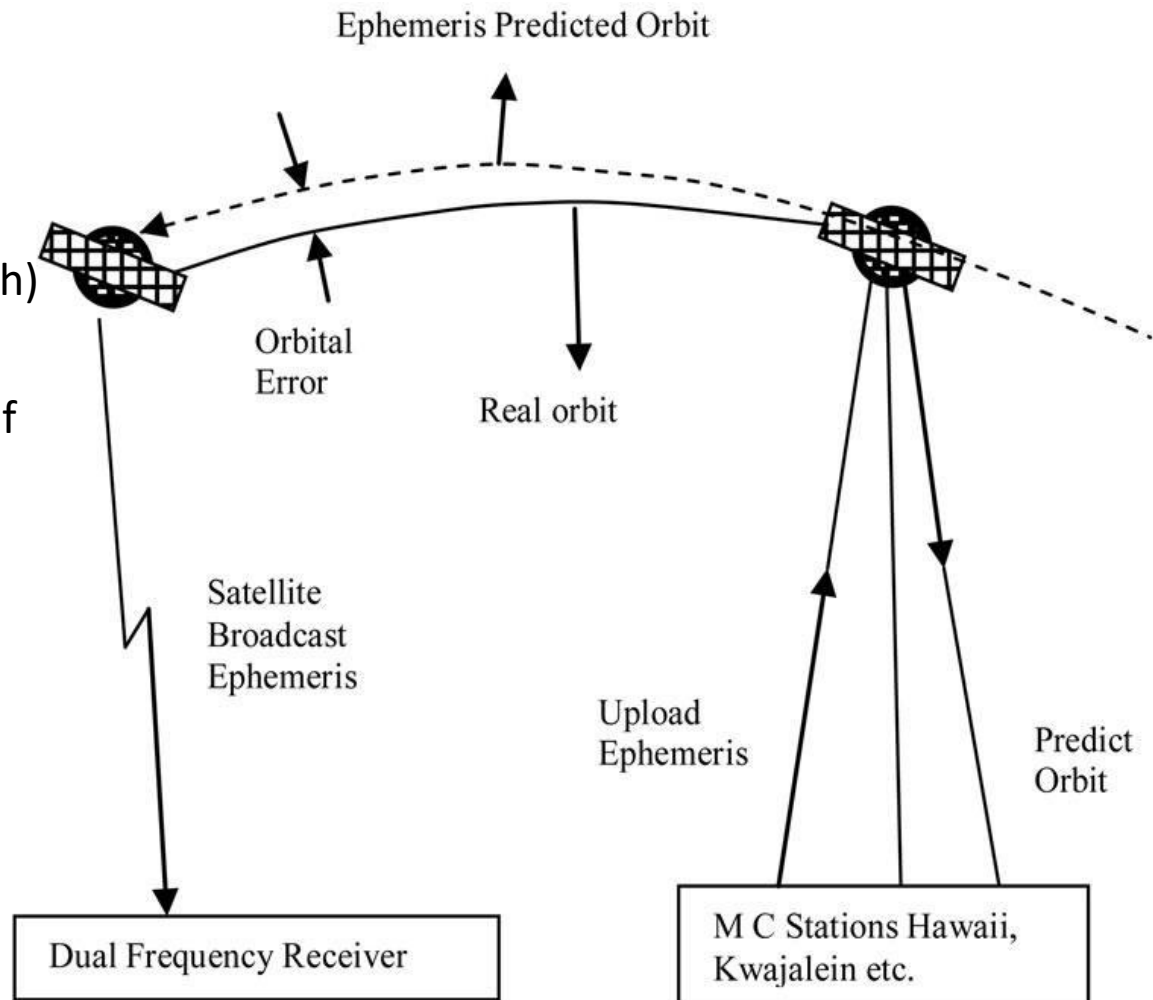


ERRESTRIAL AND MARINE PLATFORMS



SATELLITE ORBITS

- **Orbit descriptions:**
 - Ephemeris (celestial; time-dependent location)
 - Ground track (on Earth)
 - Altitude
 - Nadir
 - Snapshot area; field of view
 - >> swath



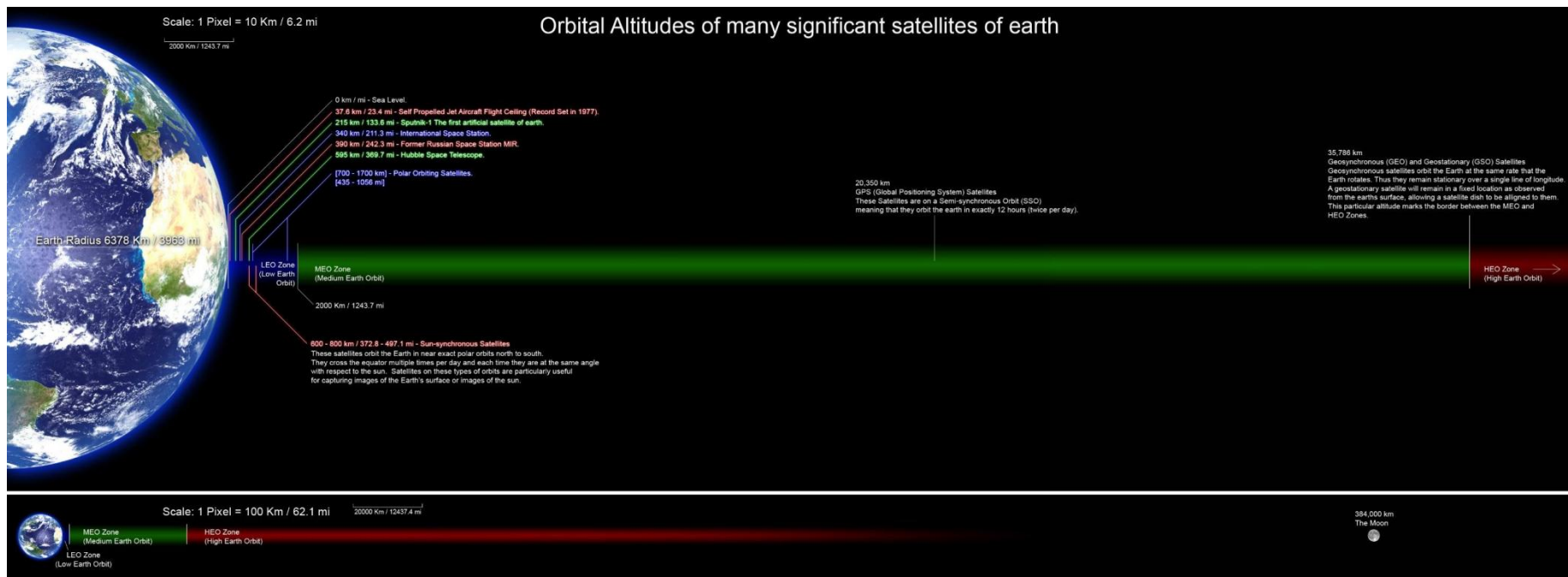
SATELLITE ORBITS

- Orbital altitude $\gg$ spacecraft speed (m/s) = $\sqrt{g \cdot m / r}$

r = orbit radius

m = mass of Earth = $5.972 \times 10^{24} \text{ kg}$

$g = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$



Scale: 1 Pixel = 10 Km / 6.2 mi

Orbital Alt

2000 Km / 1243.7 mi

Earth Radius 6378 Km / 3963 mi

0 km / mi - Sea Level.

37.6 km / 23.4 mi - Self Propelled Jet Aircraft Flight Ceiling (Record Set in 1977).

215 km / 133.6 mi - Sputnik-1 The first artificial satellite of earth.

340 km / 211.3 mi - International Space Station.

390 km / 242.3 mi - Former Russian Space Station MIR.

595 km / 369.7 mi - Hubble Space Telescope.

[700 - 1700 km] - Polar Orbiting Satellites.

[435 - 1056 mi]

LEO Zone
(Low Earth
Orbit)

MEO Zone
(Medium Earth Orbit)

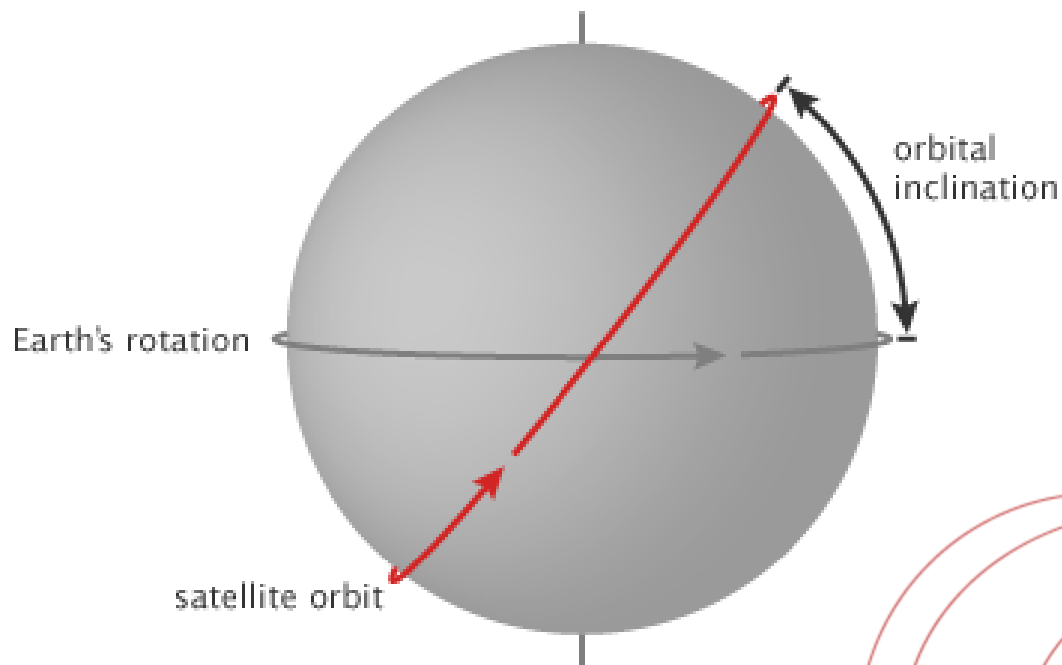
2000 Km / 1243.7 mi

600 - 800 km / 372.8 - 497.1 mi - Sun-synchronous Satellites

These satellites orbit the Earth in near exact polar orbits north to south. They cross the equator multiple times per day and each time they are at the same angle with respect to the sun. Satellites on these types of orbits are particularly useful for capturing images of the Earth's surface or images of the sun.

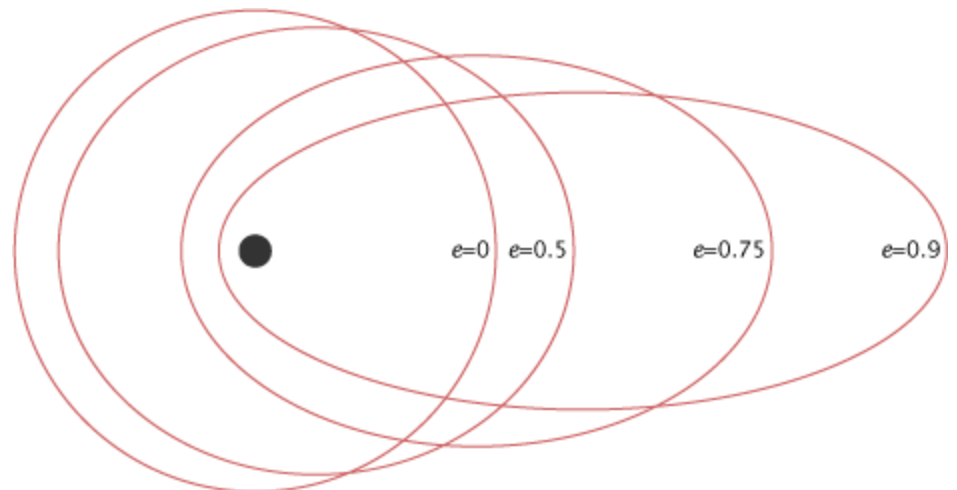
SATELLITE ORBITS

Orbital inclination angle >> coverage



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

→ $e = \sqrt{1 - b^2/a^2}$,
with a =semi-major axis, b =semi-minor axis



SATELLITE ORBITS

Orbital period: Kepler's law $T = 2\pi\sqrt{\frac{a^3}{\mu}}$
 a is the orbit's semi

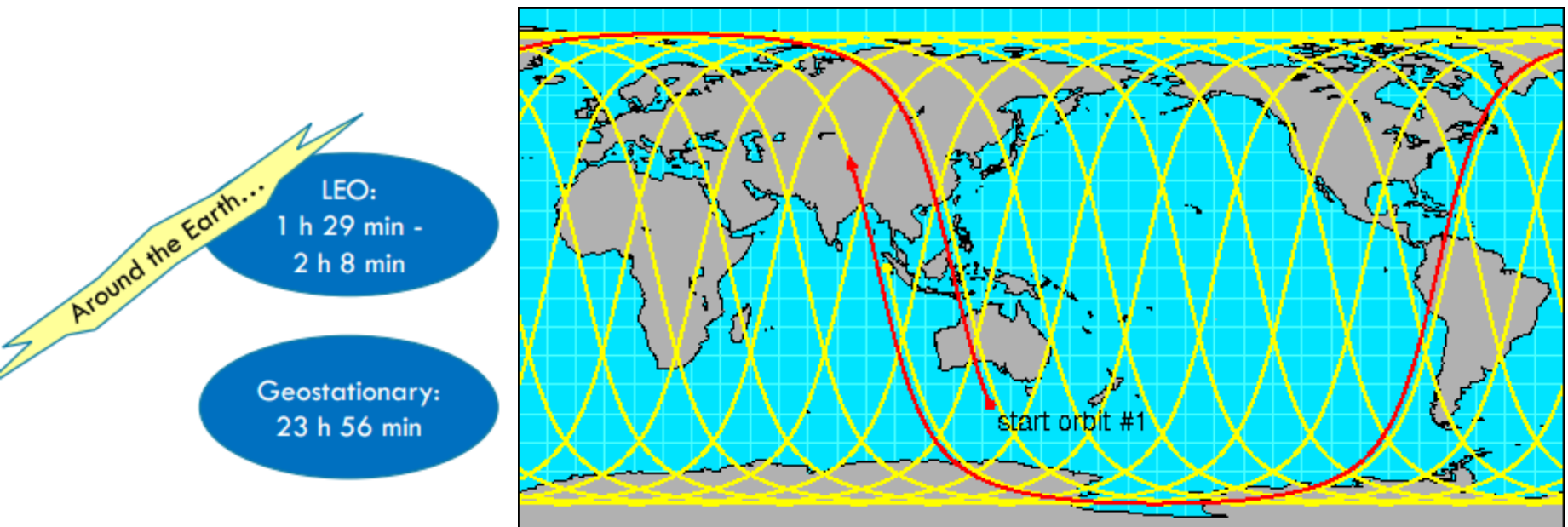
$\mu = GM$ is the standard gravitational parameter

G is the gravitational constant; $6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$

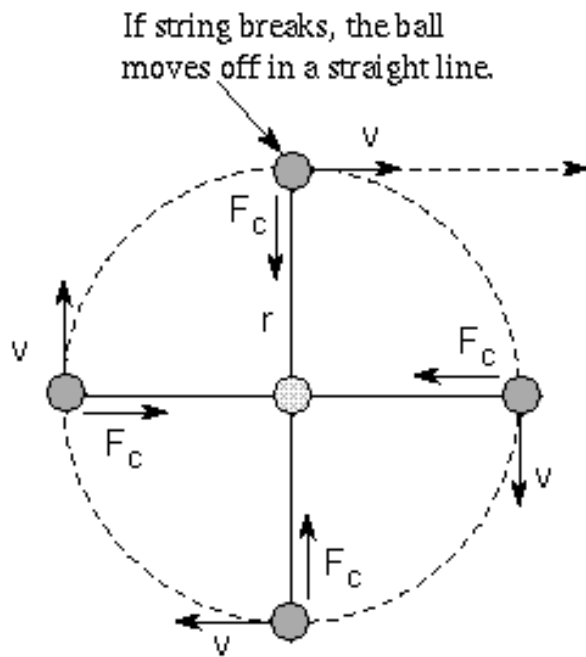
M is the mass of the more massive body (mass of Earth
 $= 5.972 \times 10^{24} \text{ kg}$).

- Repeat cycle: time between identical overpasses

<-> revisit time (repeat & pointing)



SATELLITE ORBITS



$$F_c = m v^2 / r$$

Just enough centripetal force F_c to balance the speed; just enough speed to balance the centripetal force.

$$F_g = m \cdot g_s \left(\frac{R}{r} \right)^2$$

$$F_c = m \cdot \frac{v^2}{r}, \text{ thus}$$

$$F_g = F_c \Rightarrow v = \sqrt{\frac{g_s R^2}{r}}$$

$$T = \frac{2\pi r}{v} = 2\pi r \sqrt{\frac{r}{g_s R^2}}$$

F_g : Gravitational force (N)

m : Mass of the object (kg)

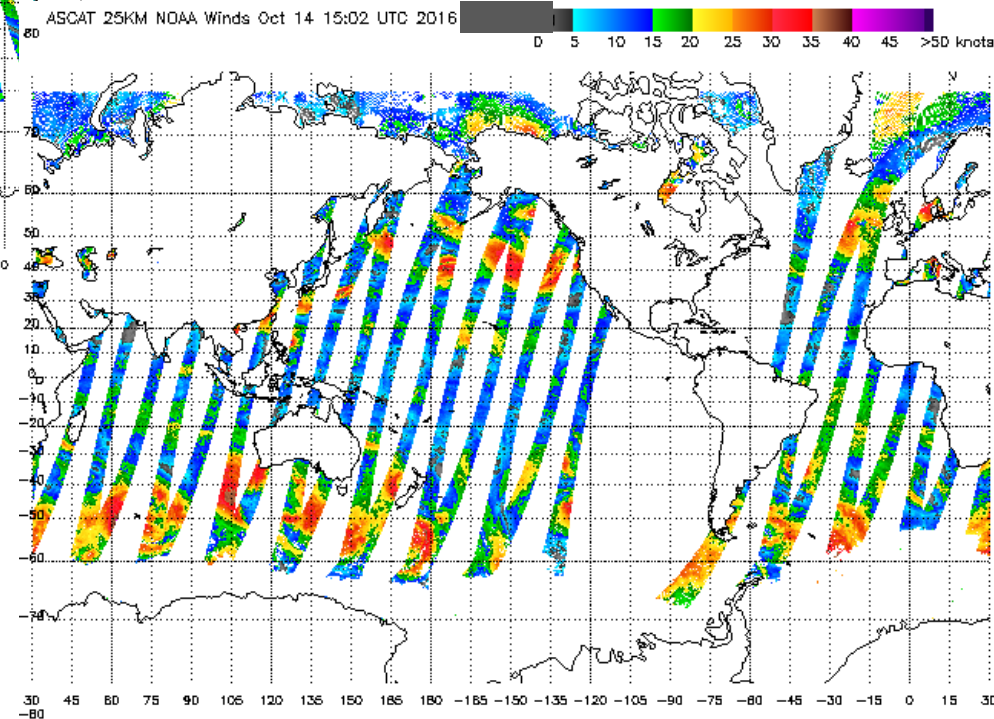
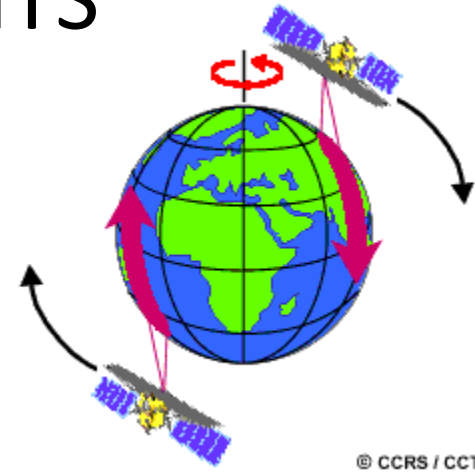
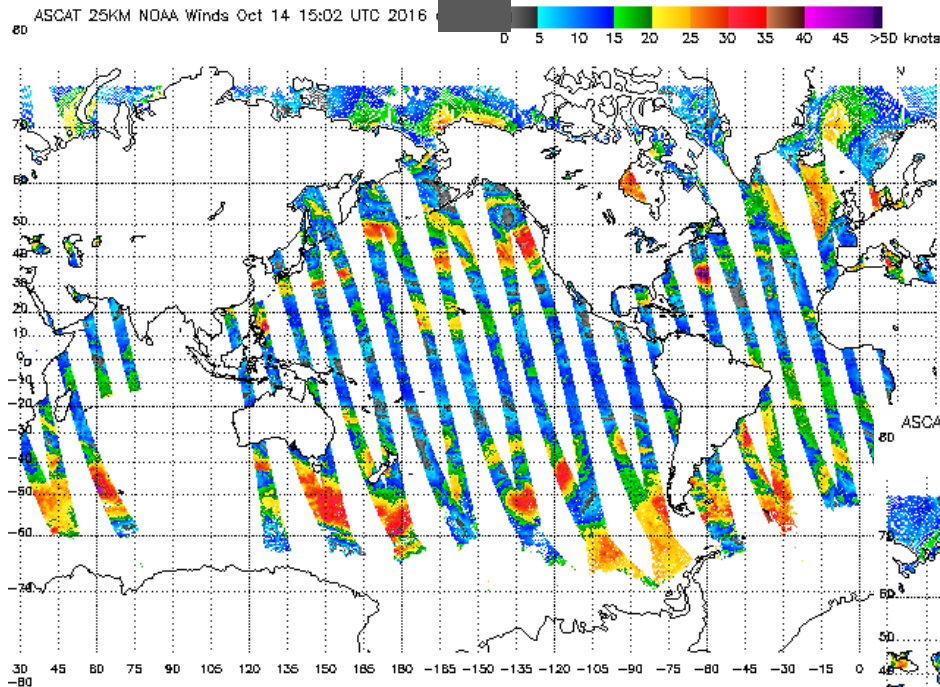
g_s : Surface gravitational acceleration (m/s²)

R : Radius of the celestial body (m)

r : Distance from the body's center to the object (m)

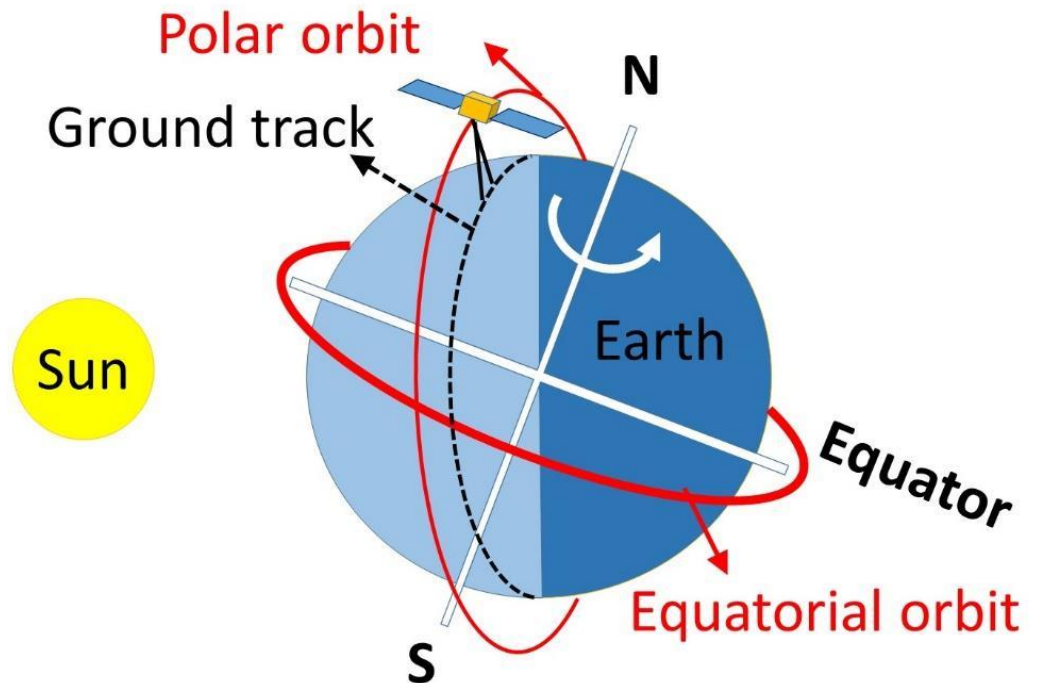
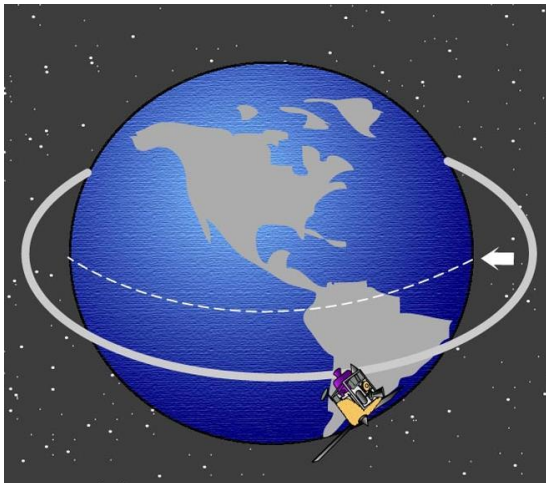
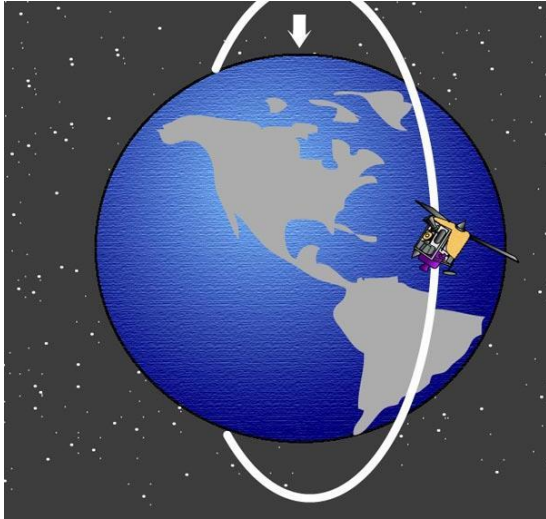
SATELLITE ORBITS

- Ascending, descending overpass



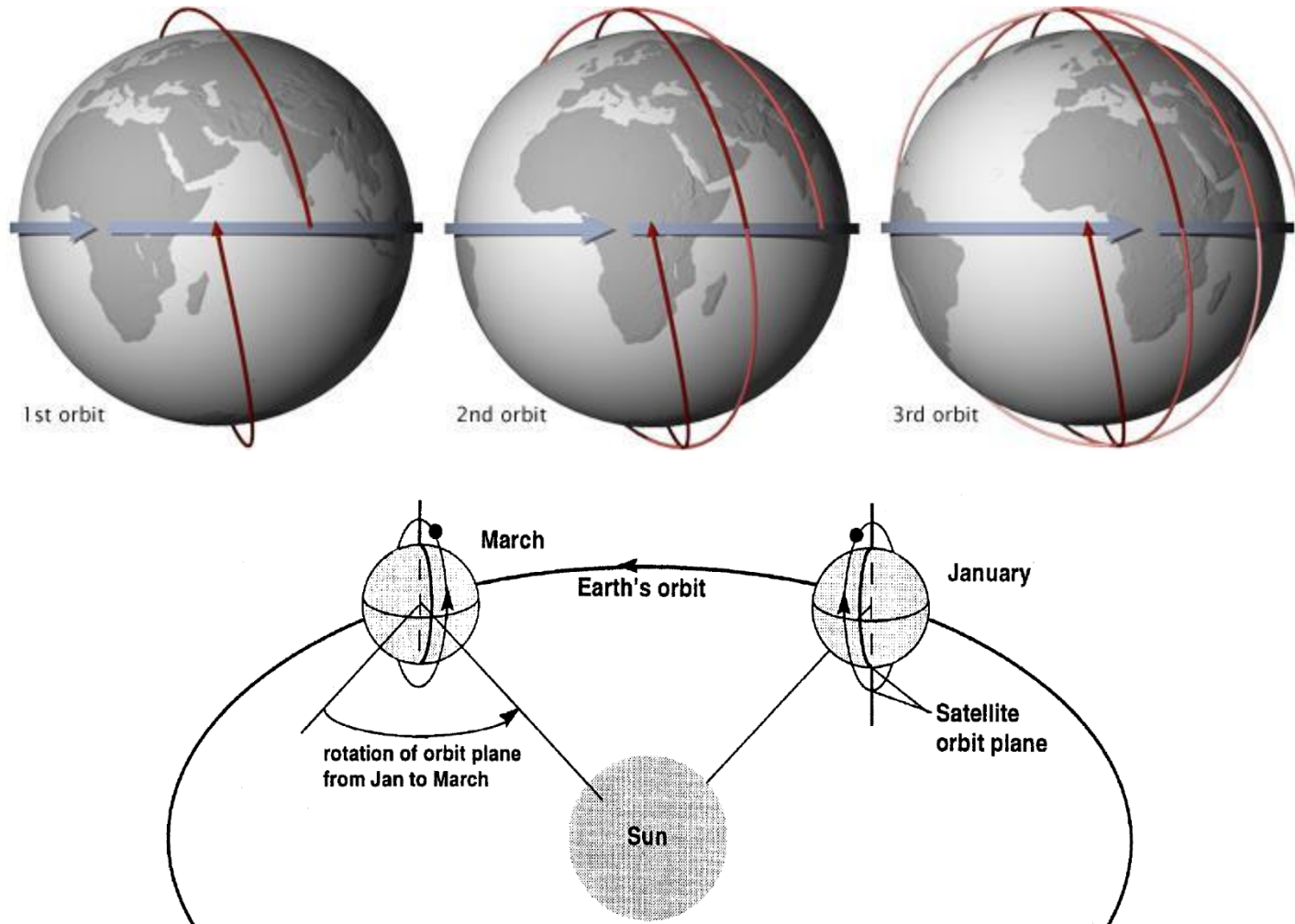
ORBIT TYPES

- Polar orbit
- Near-equatorial, low inclination orbit



ORBIT TYPES

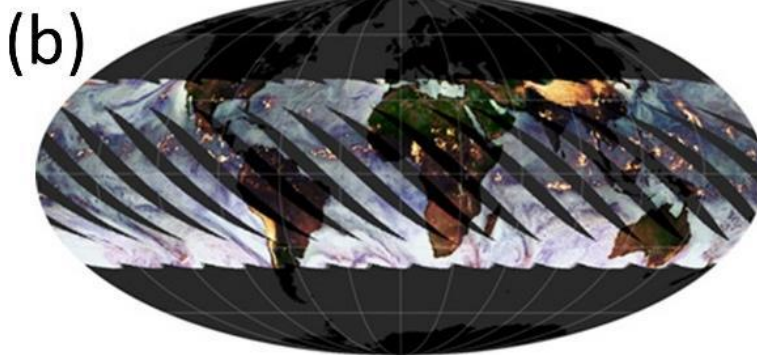
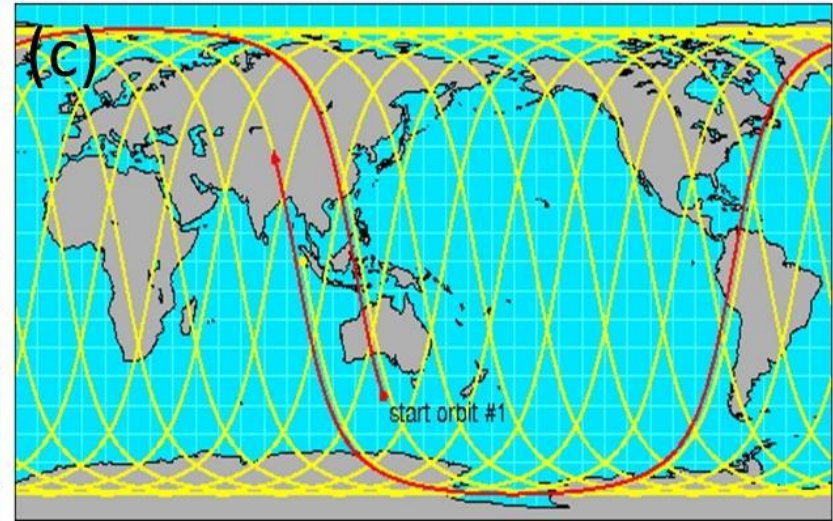
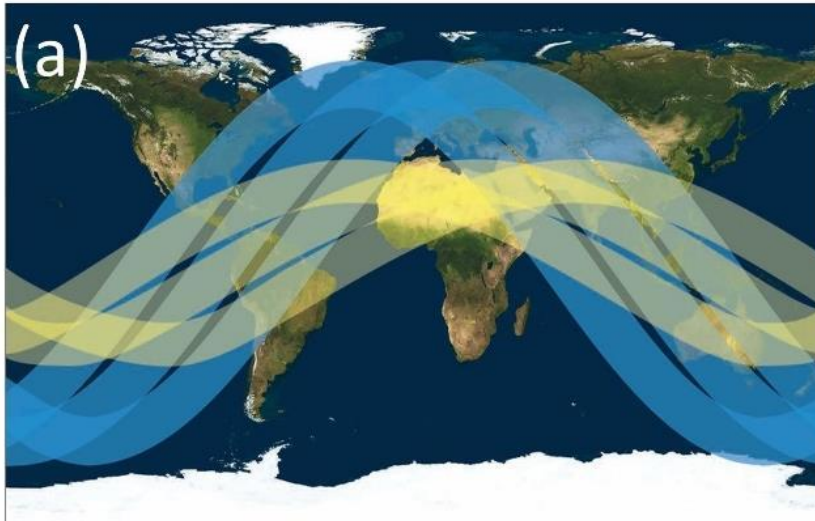
- Sun-synchronous orbit
(most polar orbits are sun-synchronous)



ORBIT TYPES

Low inclination (TRMM)
yellow, (GPM) blue

Polar orbit



Polar



Low incl



Near equatorial

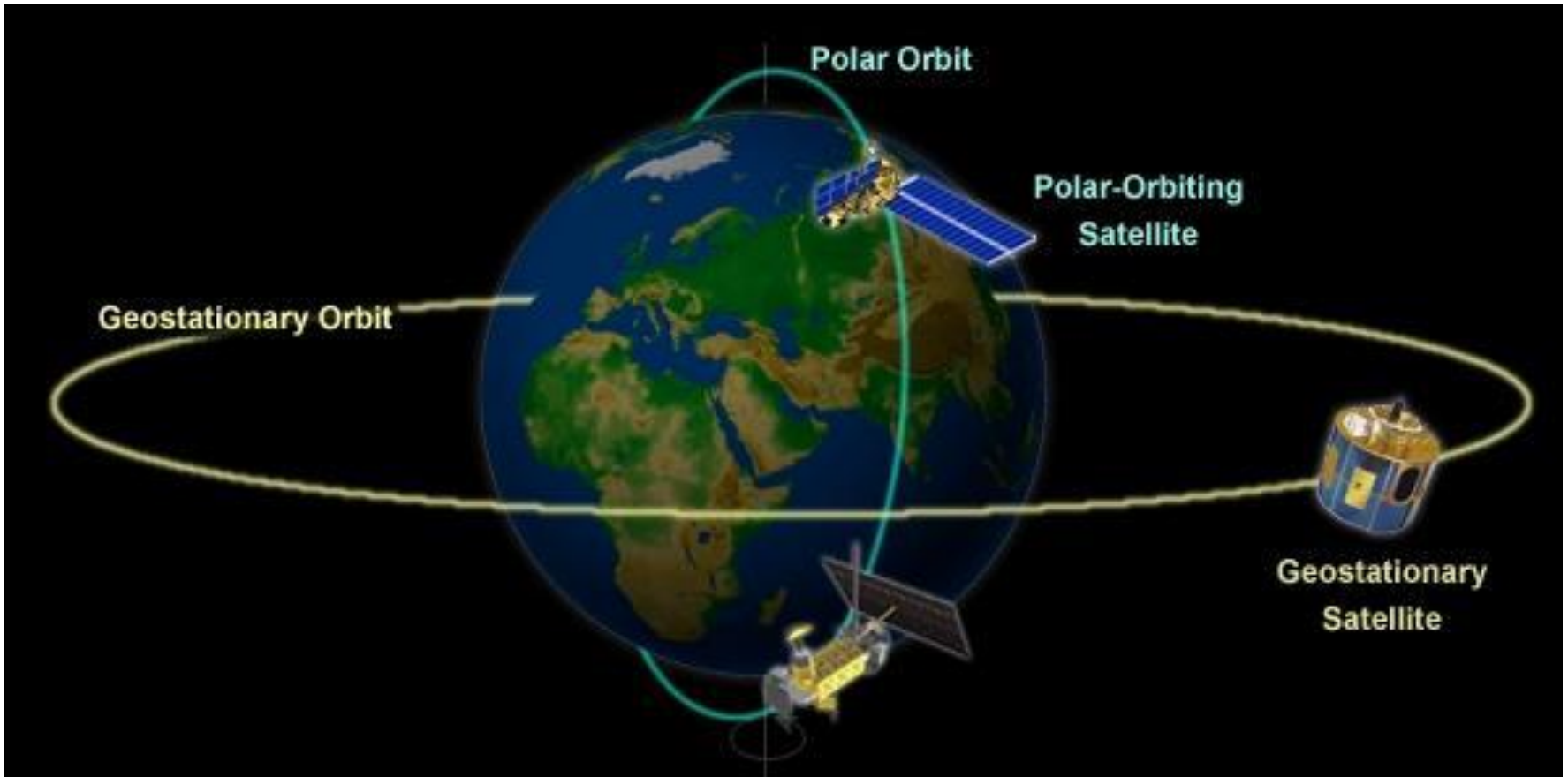
Ascending TRMM



Ascending MODIS

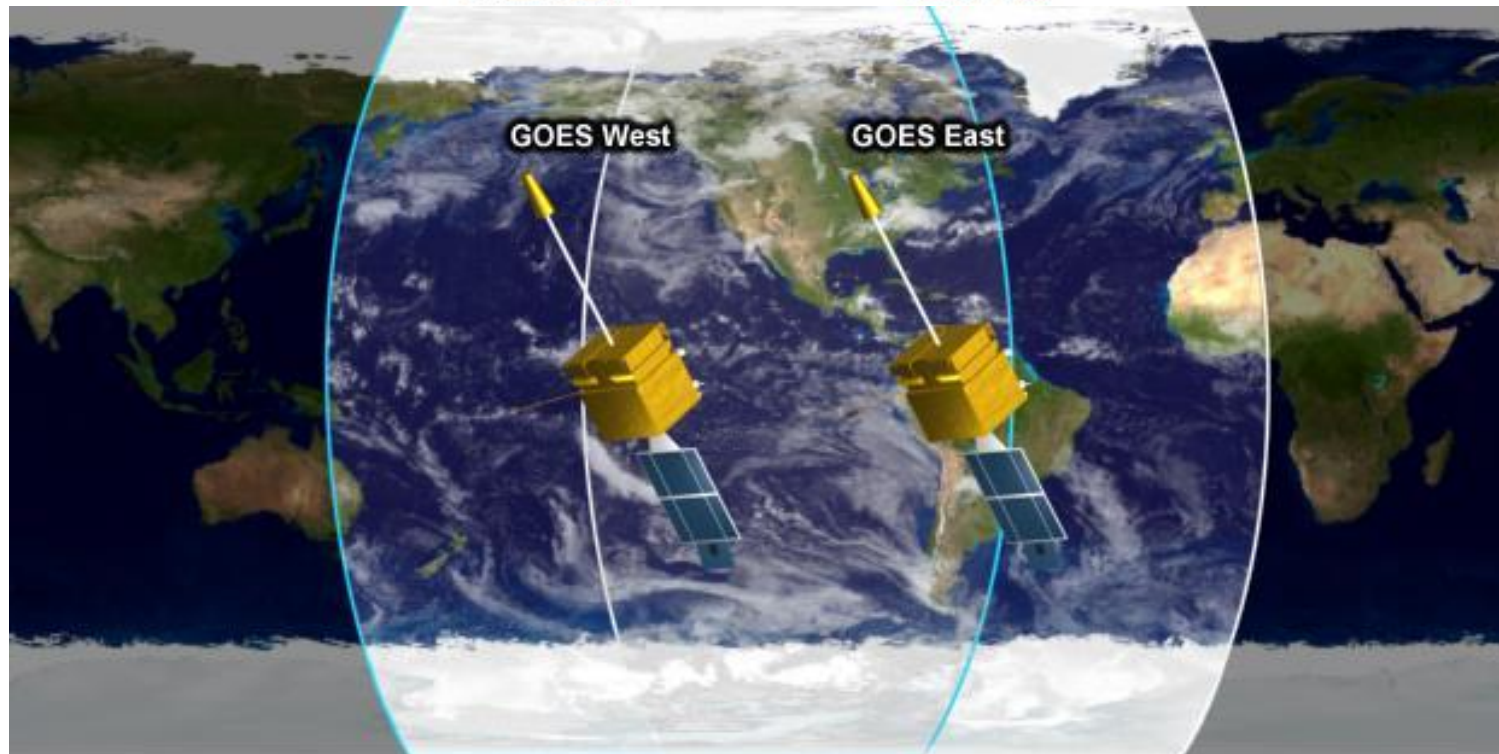
ORBIT TYPES

- Geo-stationary or geo-synchronous orbit



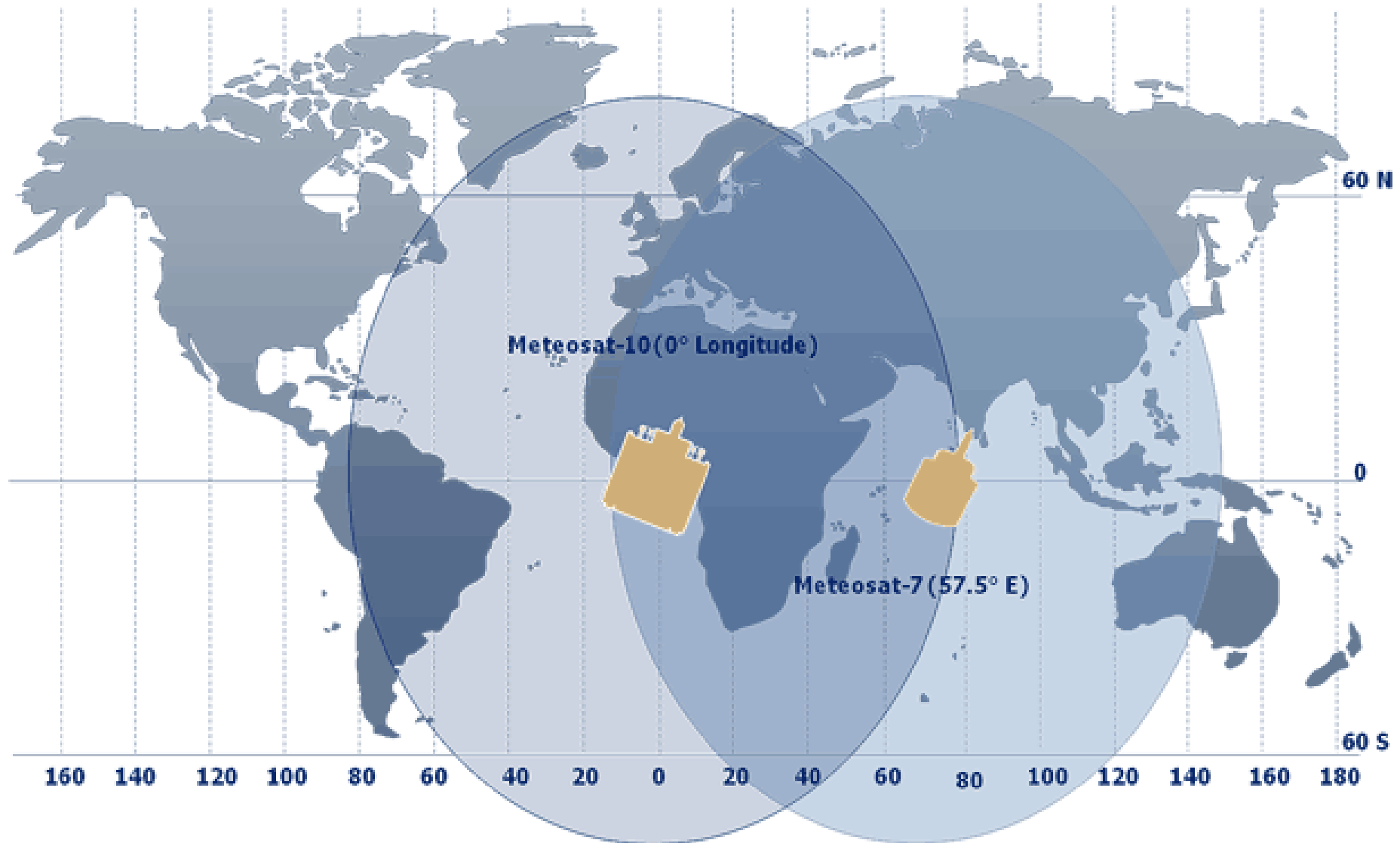
ORBIT TYPES

- Geo-stationary

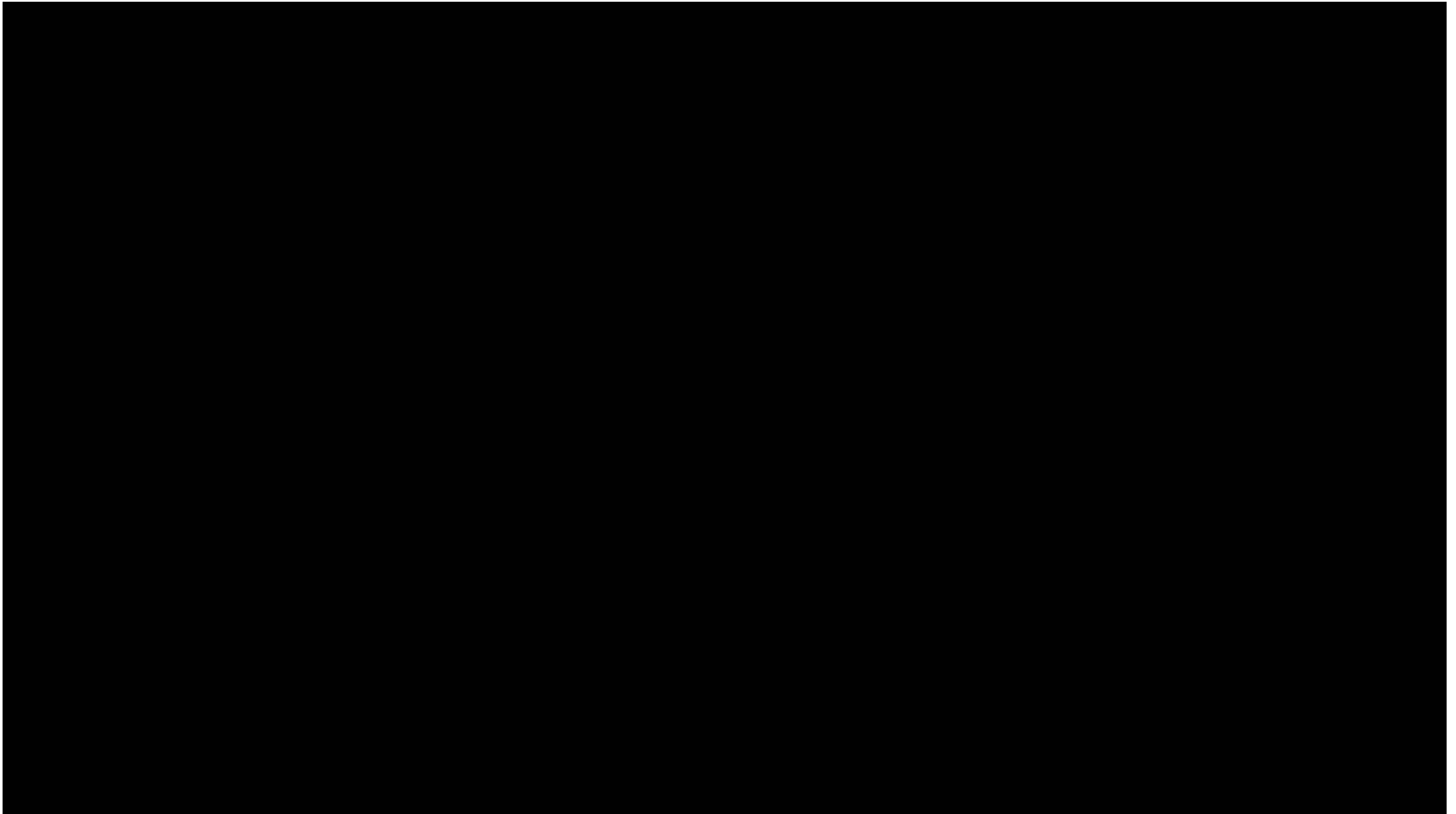


ORBIT TYPES

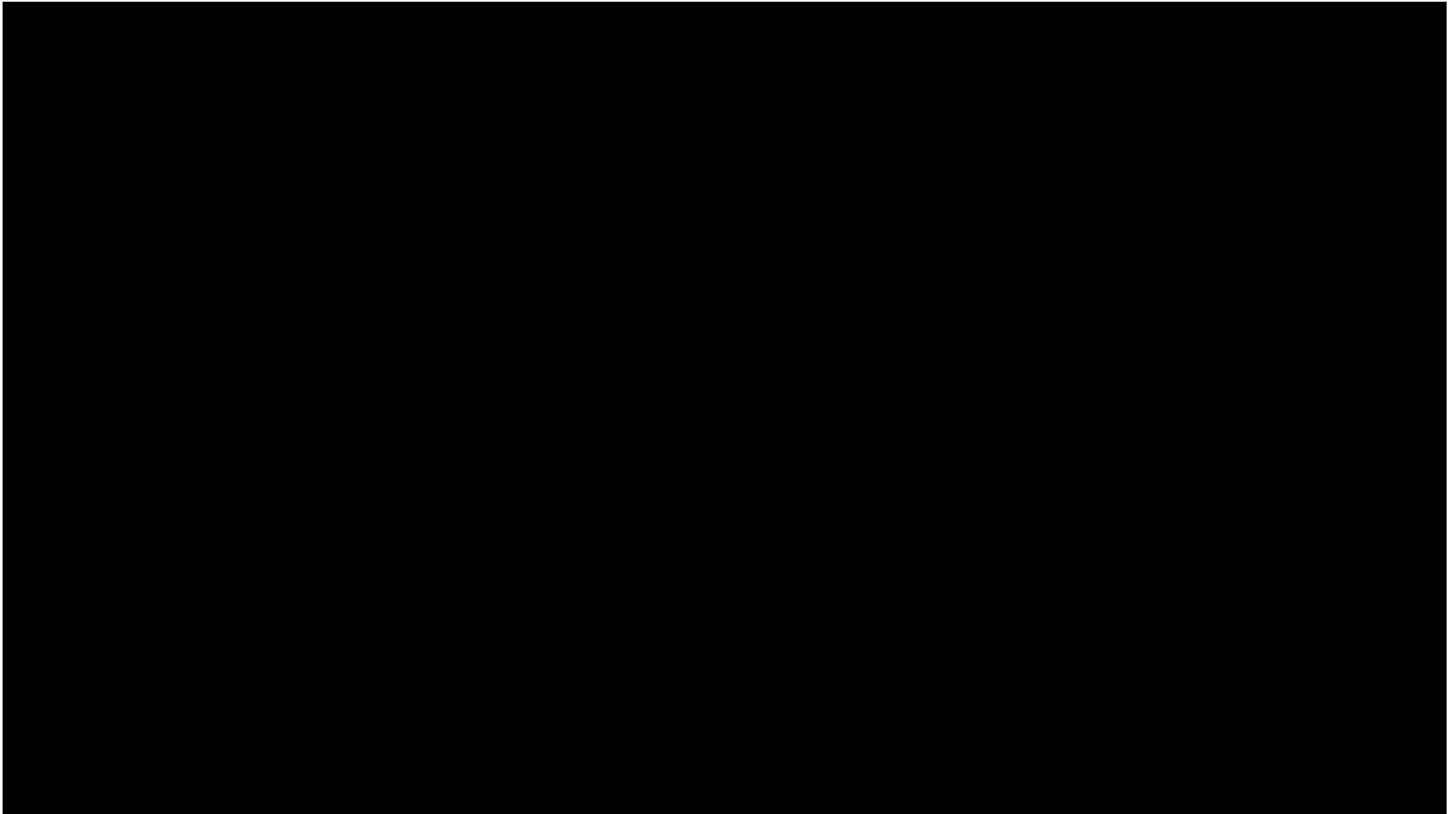
- Geo-stationary



Objects on the LEO orbits



Objects on the GEO orbits



3.4 Ground segments

- **Composition of ground station:**
 - Antenna system: transmit, receipt signals
 - Control unit: control the sensors, direction, location
 - Processing terminal: pre-processing images
 - Network equipment
 - Calibration, validation
 - Documentation

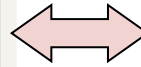
Functional Structure of the EO Ground Segment

Mission Operations

- Satellite operations and control
- Mission planning and commanding
- Monitoring
- Flight dynamics

Payload Ground Segment

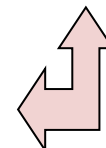
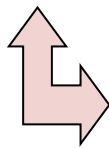
- User and service element interface
- Interface to mission planning
- Data reception and data circulation
- Processing, archiving and catalogue
- Product distribution



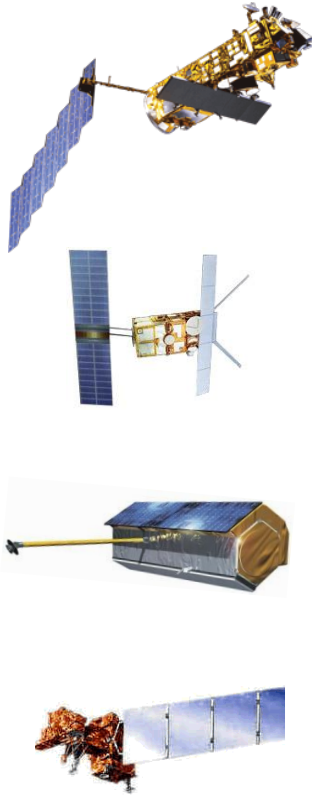
Instrument Operations and Calibration

Depending on instrument

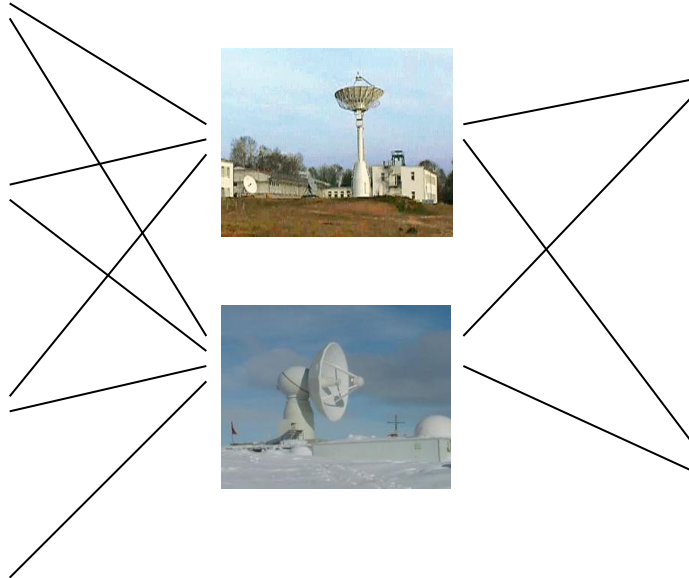
- Instrument system engineering
- Instrument operations
- Calibration and long term monitoring



Missions, Facilities and Users



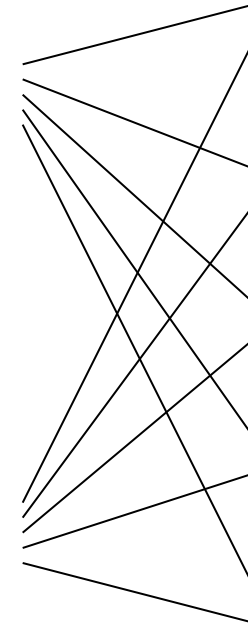
TerraSAR-X
ERS-2, ENVISAT
IRS-Series
CHAMP, GRACE



Receiving, controlling
Stations

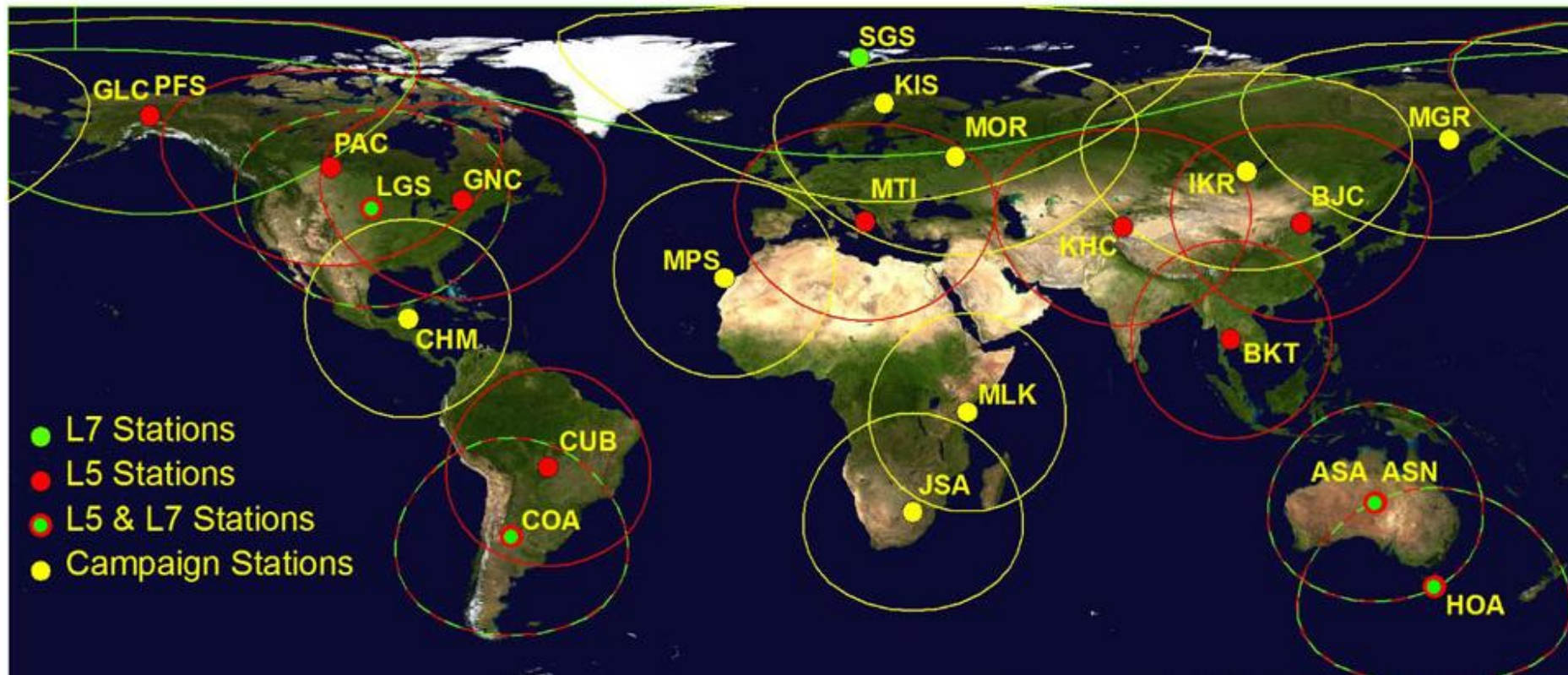


Processing
Archiving, Distribution

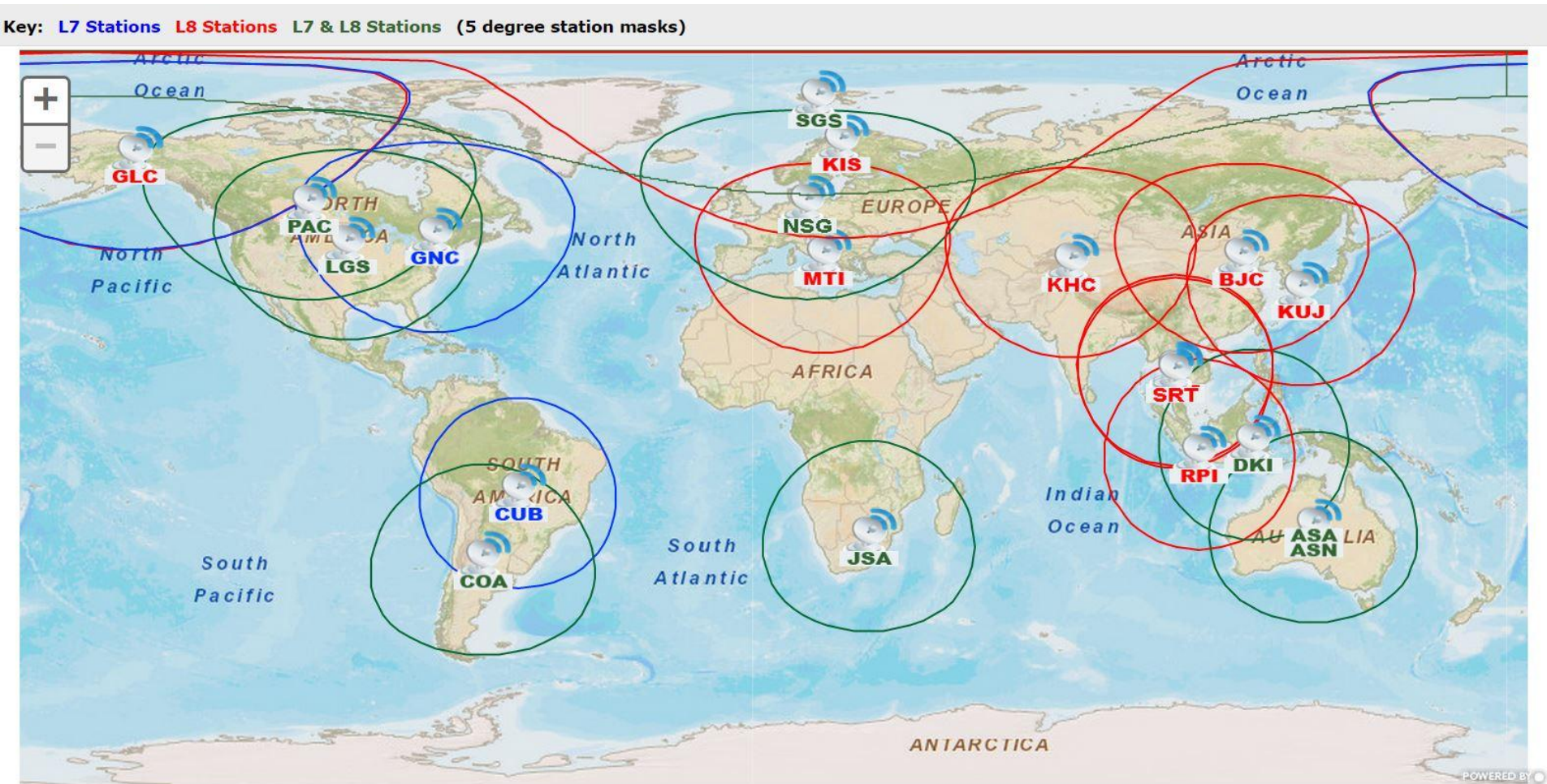


Service-Elements
& Users

Active Landsat Ground Stations

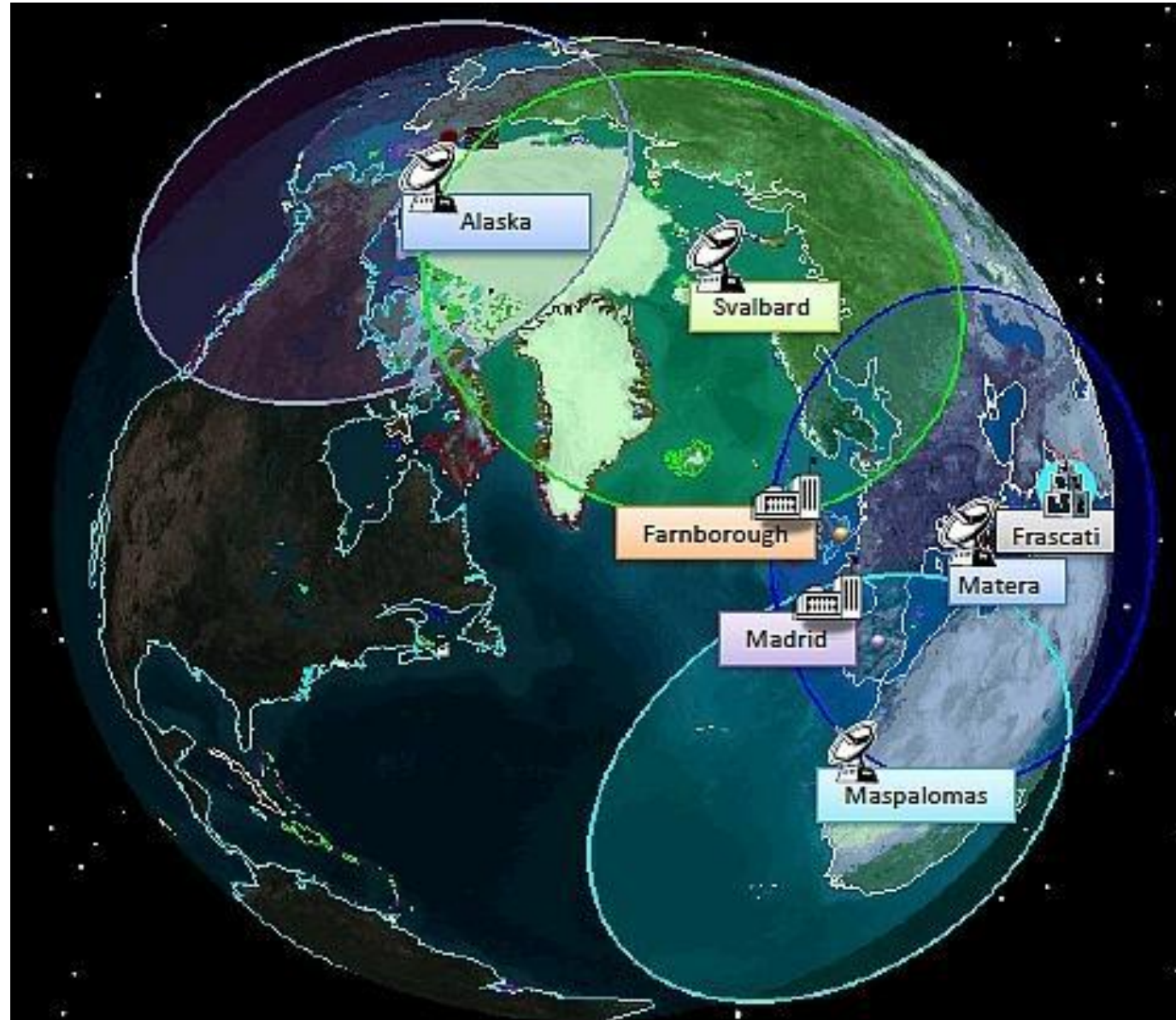


Landsat 7 and Landsat 8 ground stations

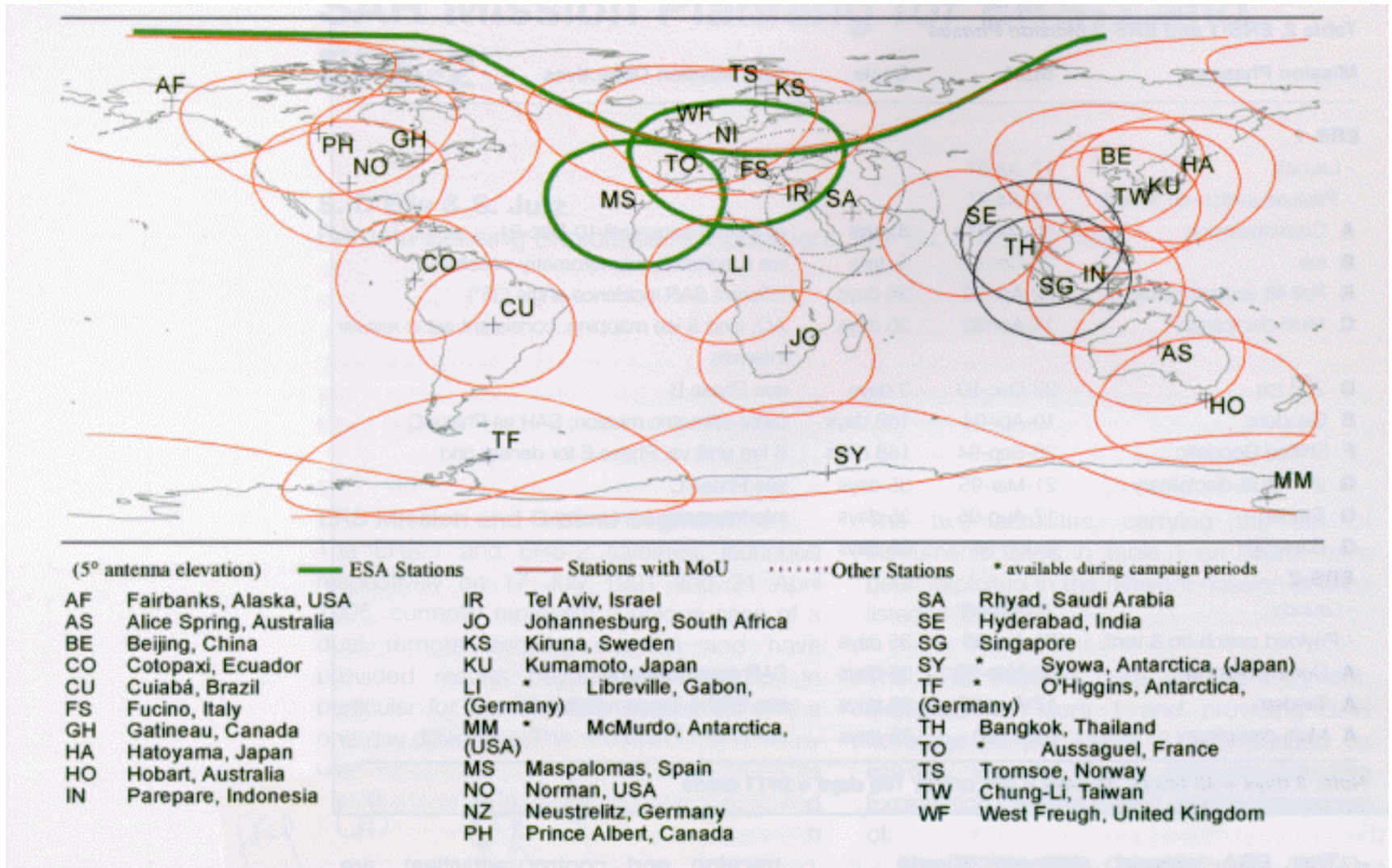


Payload Data Ground Segment Sentinel-2

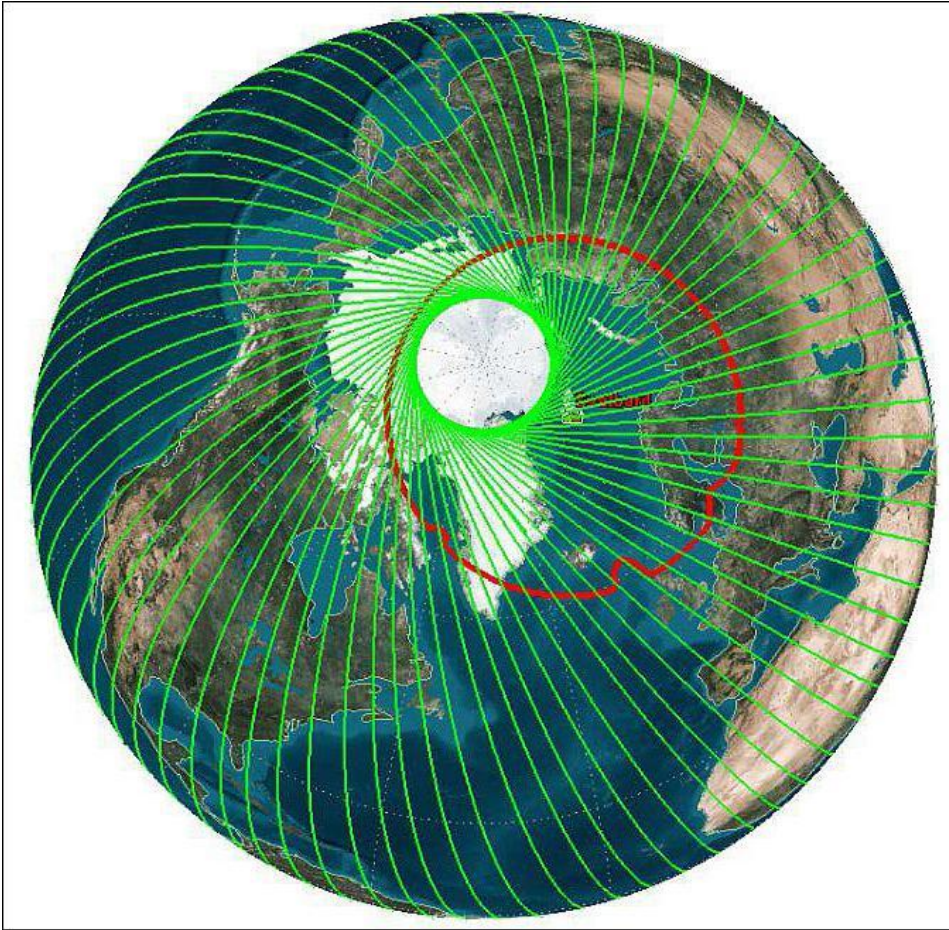
- CGS (Core Ground Stations: Matera (Italy), Maspalomas (Spain), Svalbard (Norway), Alaska (USA).
- PAC (Processing/Archiving Center): Farnborough (UK), Madrid (Spain)
- MPC (Mission Performance Center): TBD
- PDMC (Payload Data Management Center): ESA/ESRIN, Frascati, Italy.



Ground stations for ERS-1



SVALSAT Satellite Ground Station



RapidEye constellation (image credit: BlackBridge)



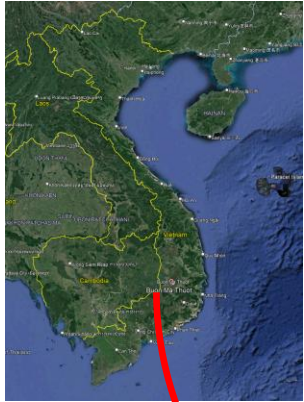
SvalSat station (Norwegian) in Antarctica, the only ground stations able to download information for all 14 orbits of polar orbiting satellites

Ground segments



Calibration devices of the DLR SAR

Ground segments



*Calibration area for
VNRedsat 1
satellite, in Dak
Lak, VN*



*Calibration sample at Eglin Air
Force Base, Florida*

Thank you for your attention!